Is Generative AI Removing the First Rung?

Automation Exposure and Entry-Level Job Ads in Europe

Tomáš Oleš*

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Abstract

Generative AI performs many of the tasks of entry-level jobs. I ask whether it removes entry-level jobs in Europe, with 246 million online job ads posted in 28 European countries between 2020 and 2025 and with automation exposure measured by matching breakthrough patents to the tasks of four-digit occupations. I compare occupations in the top fifth and in the bottom fifth of automation exposure to natural language processing, within countries, before and after the release of ChatGPT. In the top fifth, entry-level ads fell by 24 log points relative to the bottom fifth, and senior ads fell by 20 log points, so the entry-level share did not fall. What changed is what entry-level jobs require. Per standard deviation of automation exposure, the share of entry-level ads that require language, digital, analytical, social, interpersonal, and people-management skills fell by 1.0 to 1.4 percentage points, while manual skill requirements did not change. The declines appeared in 2024 and grew in 2025, and analytical and people-management skill requirements fell more in entry-level ads than in senior ads. Exposed occupations also asked less often for tertiary education and more often for mid-level education, which includes vocational education, but skill requirements fell within each education level as well. Exposure to industrial robots predicts none of the changes in skill requirements. The decline in the number of job ads is not present among job ads from the same sources, whereas the change in skill requirements is.

Keywords: generative AI; entry-level jobs; automation; skill requirements; online job ads
JEL: J23, J24, O33

1 Introduction

When AI can perform the tasks of entry-level jobs, does it remove entry-level jobs? In many occupations, the tasks of entry-level jobs are codifiable: drafting text, summarizing documents, writing routine code, answering standard questions. Generative AI performs these tasks well. If firms give the tasks of entry-level jobs to AI, fewer workers acquire the experience that senior jobs require.

The first evidence from the United States suggests that this is happening. Brynjolfsson et al. (2025a) show in payroll data that employment of workers aged 22 to 25 fell in the occupations most exposed to generative AI, relative to the least exposed occupations, while employment

*Bratislava University of Economics and Business, Faculty of Economics and Finance. tomas.oles@euba.sk.

of older workers in the same occupations did not fall. Hosseini and Lichtinger (2025) find that firms that adopted generative AI reduced hiring into entry-level jobs relative to senior jobs. Experimental studies point the other way: AI raises the productivity of less experienced workers the most (Noy and Zhang, 2023; Brynjolfsson et al., 2025b; Dell'Acqua et al., 2023), although in a field experiment with patent lawyers the gains of less experienced lawyers did not persist once they worked without AI (Autor et al., 2026).

This paper asks whether generative AI removes the first rung in Europe, and how it changes what the first rung requires. I use the online job ads collected by Eurostat's Web Intelligence Hub (WIH) in the 27 member states of the European Union and the United Kingdom between 2020 and 2025. By an *entry-level ad*, I refer to a job ad requiring no experience, or experience less than a year, and by a *senior ad*, to a job ad requiring more than a year of experience. Job ads have two advantages over employment data. They record the experience requirement directly, so no age threshold stands in for experience, and they record the skill requirements of each job, so I observe not only how many entry-level jobs firms offer but also what these jobs require. I measure the *automation exposure* of each four-digit occupation by matching the texts of breakthrough patents granted before 2016 to the tasks that the occupation's workers perform, separately for each AI type in the USPTO AI Patent Dataset (Pairolero et al., 2024). I compare occupations in the most and the least exposed fifth, within countries, before and after the release of ChatGPT in November 2022, with entry-level ads in the role of young workers and senior ads in the role of older workers.

My first finding is that generative AI did not remove the first rung more than the rungs above it. In the top fifth of automation exposure to natural language processing, firms posted 24 log points fewer entry-level ads after 2022 than in the bottom fifth, but they also posted 20 log points fewer senior ads. The entry-level share therefore did not fall. Unlike in the United States, where the decline is specific to young workers, exposed occupations in Europe offered fewer jobs at every level of experience.

My second finding is that generative AI changed what entry-level jobs require. Per standard deviation of automation exposure, the share of entry-level ads that require language, digital, analytical, social, interpersonal, and people-management skills fell by 1.0 to 1.4 percentage points, and entry-level ads listed about 5 percent fewer skill requirements. Manual skill requirements did not change. The declines appeared in 2024, the second year after the release of ChatGPT, and grew in 2025, when digital and language skill requirements in entry-level ads were about 2 percentage points per standard deviation below their 2022 level. These are the skill types that large language models now supply.

My third finding is that entry-level ads lost analytical and people-management skill requirements faster than senior ads. In terms of a simple task model, this suggests that generative AI automates the analytical and coordinating tasks of entry-level jobs more than proportionally, while it automates language and digital tasks in entry-level and senior jobs alike. Exposure to industrial robots, the placebo technology, predicts none of these changes. The declines in skill requirements hold within industries, without the United Kingdom, without ICT occupations, and among job ads from the same sources, where they are smaller. The faster loss in entry-level ads is less robust: the difference keeps its sign in every specification for people-management

skill requirements and in all but one for analytical skill requirements, but it is often imprecise. The decline in the number of job ads holds within industries, without the United Kingdom, and without ICT occupations, but not among job ads from the same sources.

My fourth finding concerns education. In exposed occupations, senior ads asked less often for tertiary education and more often for mid-level education, the upper secondary and post-secondary level that includes vocational education and apprenticeships, and entry-level ads asked more often for mid-level education and less often for low education. Entry-level ads that require mid-level education fell less than entry-level ads that require tertiary or low education. The shift in education requirements does not explain the change in skill requirements: entry-level ads lost skill requirements at every education level. In job ads that require mid-level education, entry-level ads and senior ads lost the same skill types; in job ads that require tertiary education, entry-level ads lost more social, people-management and analytical skill requirements than senior ads, although this difference is imprecise.

The paper contributes to two literatures. It adds European evidence to the literature on generative AI and entry-level jobs, which so far relies on data from the United States, and it shows that the first effect of generative AI on the first rung is on its content rather than on its size. It also extends the literature on skill requirements in job ads (Deming and Kahn, 2018; Hershbein and Kahn, 2018; Oleš, 2026) from differences across occupations to changes within occupations around a large technological shock. In nearly all online job ads posted in Slovakia in 2022, Oleš (2026) shows that skill requirements differ systematically with automation exposure: job ads that require people-management, software-specific and financial skills belong to less exposed occupations, and job ads that require routine cognitive, customer-service and social skills to more exposed ones. A single cross-section, however, cannot show how skill requirements change within occupations when a new technology arrives. The closest evidence on such changes comes from Germany. Linking the job ads of the Federal Employment Agency to establishment data, Peede and Stops (2026) find that establishments with a larger share of job ads that require AI skills between 2015 and 2023 reduced their demand for other skills, mostly skills required for non-routine analytical tasks, with no association with employment growth. My results point in the same direction with a different measure, automation exposure rather than AI adoption, and around the release of ChatGPT: in exposed occupations, the skill requirements of entry-level ads fell, analytical skill requirements among them, while the decline in the number of job ads depends on the sources. Section 2 presents the framework, Section 3 the data and the measures, and Section 4 the empirical strategy. Section 5 presents the results, and Section 6 discusses them and concludes.

2 Framework

This section presents a stylized task model in the spirit of Acemoglu and Restrepo (2018) and Autor et al. (2024). It shows what the response of entry-level ads and senior ads to automation reveals about the tasks that AI automates.

Consider an occupation that produces output y by combining tasks $i \in [0, N]$ with unit elasticity, so the occupation spends the same amount, $p\,y/N$, on every task, where p is the price of its output. Capital performs the automated tasks, a set of measure A. Workers perform the

remaining tasks in entry-level jobs and in senior jobs. Let T_e be the measure of *entry-level tasks*, the tasks performed in entry-level jobs, and T_s the measure of *senior tasks*, with $T_e + T_s = N - A$, and let $\sigma = T_e/(T_e + T_s)$ be the share of entry-level tasks among the tasks that workers perform. With wages w_e and w_s, the occupation's demand for entry-level jobs is $L_e = T_e\, p\, y/(N\, w_e)$ and its demand for senior jobs is $L_s = T_s\, p\, y/(N\, w_s)$.

Automation raises A. Let α be the share of entry-level tasks among the tasks that AI automates, so that $dT_e = -\alpha\, dA$ and $dT_s = -(1-\alpha)\, dA$. Holding output and wages fixed, the number of entry-level jobs and the number of senior jobs change by

$$\frac{d\ln L_e}{dA} = -\frac{\alpha}{T_e}, \qquad \frac{d\ln L_s}{dA} = -\frac{1-\alpha}{T_s}, \tag{1}$$

and an increase in output raises both by the same proportion.

Proposition 1 (Jobs)**.** *Automation lowers the number of entry-level jobs relative to the number of senior jobs, $d\ln(L_e/L_s)/dA < 0$, if and only if AI automates entry-level tasks more than proportionally, $\alpha > \sigma$. If $\alpha = \sigma$, entry-level jobs and senior jobs fall by the same proportion and the entry-level share does not change.*

The same allocation of tasks determines the skill requirements of jobs. Each task requires a set of skills, and a job ad lists the skill requirements of the tasks of the job. Let r_e^g be the share of entry-level tasks that require skill type g, and a^g the share of the automated entry-level tasks that require it; define r_s^g and b^g in the same way for senior tasks. Then

$$\frac{dr_e^g}{dA} = -\frac{\alpha}{T_e}\,(a^g - r_e^g), \qquad \frac{dr_s^g}{dA} = -\frac{1-\alpha}{T_s}\,(b^g - r_s^g). \tag{2}$$

Proposition 2 (Skill requirements)**.** *Automation lowers the share of entry-level jobs that require skill type g if and only if AI automates the entry-level tasks that require g more than proportionally, $a^g > r_e^g$. It lowers this share more in entry-level jobs than in senior jobs if and only if $\frac{\alpha}{T_e}(a^g - r_e^g) > \frac{1-\alpha}{T_s}(b^g - r_s^g)$.*

The propositions do not sign the effects; the data do. Proposition 1 links the number of entry-level ads and senior ads to the share of entry-level tasks among the automated tasks. Proposition 2 links the skill requirements of entry-level ads and senior ads to the skill content of the automated tasks. The difference between entry-level ads and senior ads, in the number of job ads and in skill requirements, is a triple difference: it compares experience levels within occupations, occupations with high and low automation exposure, and the years before and after the release of ChatGPT. Industrial robots provide a placebo. They automate tasks, but nothing like the release of ChatGPT changed what industrial robots can do in 2022 or 2023, so occupations with high exposure to industrial robots should show no change after 2022.

3 Data and measurement

3.1 Online job ads

The data are the online job ads collected by Eurostat's Web Intelligence Hub. The WIH scrapes job portals, public employment services and company websites, removes duplicate job ads, and

codes each job ad by occupation (four-digit ISCO-08), industry (NACE Rev. 2) and location. The WIH also extracts the experience requirement and the ESCO skills that each job ad mentions. The experience requirement is coded in bands: no experience, less than one year, one to two years, two to four years, four to six years, and more than six years. An entry-level ad is a job ad in the first two bands; a senior ad is a job ad in the other bands.

I use all job ads posted between 2020 and 2025 in the 27 member states of the European Union and the United Kingdom. A *cell* is a combination of country and four-digit occupation. I keep the cells with at least ten job ads that state an experience requirement in every year, which gives a balanced panel of 7,203 cells in 411 occupations and 28 countries, with 246 million job ads, of which 130 million state an experience requirement. In the pre-period, the years 2020 to 2022, these cells posted 11.6 million entry-level ads and 10.1 million senior ads per year. By the *entry-level share*, I refer to entry-level ads as a share of job ads that state an experience requirement; it was 53.3 percent in the pre-period. The post-period is the years 2023 to 2025.

Comparisons across countries would mislead. The entry-level share of all job ads fell from 53.8 percent in 2022 to 47.1 percent in 2025 and 43.9 percent in the first half of 2026 (Appendix Table B.1). With the country composition of job ads held at its 2021–22 level, however, the entry-level share stayed between 53 and 54 percent from 2021 to the first half of 2026. The aggregate decline reflects the falling share of the United Kingdom, where the entry-level share is high, in the job ads that the WIH collects: from 30 percent of job ads that state an experience requirement in 2022 to 10 percent in 2025. All comparisons below are therefore made within countries. Appendix A documents the job ads by country, the coding of the experience requirement, the sources, and the two countries whose data changed most, the United Kingdom and Poland.

3.2 Skill requirements

Skill requirements are measured in entry-level ads and in senior ads. A *skill requirement* is a distinct ESCO skill in a job ad. Each of the 13,989 skills, knowledge items and transversal competences in ESCO belongs to at most one *skill type*, assigned by rules applied in a fixed order (Appendix Table B.2). Language skills are the skills that ESCO flags as languages and the groups for using and mastering languages. Digital skills are working with computers, the skills that ESCO flags as ICT, and working with digital devices and applications. Analytical skills are solving problems, conducting studies and investigations, calculating and estimating, analyzing information and data, and the transversal thinking skills. Social skills are negotiating, liaising and networking, presenting information, working with others, communicating, and collaborating. Interpersonal skills are teaching and training, advising and consulting, promoting and selling, counseling, and supporting the public, clients and others. People-management skills are leading and motivating, building teams, recruiting, and supervising people. Writing skills are writing, drafting, editing and proofreading. Manual skills are handling and moving, constructing, working with machinery, and physical skills. The skill types simplify the classification of Oleš (2026).

The skill data are available for three-digit occupations. For skill requirements, a cell is therefore a combination of country and three-digit occupation, with at least ten entry-level ads and ten senior ads in every year: 2,516 cells in 122 occupations and 28 countries, with 67 million entry-level ads and 65 million senior ads. For each cell, year and experience level, I measure

the share of job ads that require at least one skill of each skill type and the log number of skill requirements per job ad. In the pre-period, 67 percent of entry-level ads required digital skills, 48 percent analytical skills, 35 percent people-management skills, 34 percent language skills, 19 percent manual skills, and 2 percent writing skills.

3.3 Education requirements

The WIH codes the education requirement of a job ad in ISCED levels. I group them into low education (ISCED 0 to 2), mid-level education (ISCED 3 and 4: upper secondary and post-secondary non-tertiary education, which includes vocational education and apprenticeships), and tertiary education (ISCED 5 to 8). The education requirement is stated in 98 percent of job ads; it is missing for most job ads only in Denmark and Portugal (Appendix Table A.1). For education, a cell is a combination of country and four-digit occupation with at least ten entry-level ads and ten senior ads that state an education requirement in every year: 4,853 cells in 396 occupations. In the pre-period, 50 percent of entry-level ads in these cells required mid-level education, 45 percent tertiary education, and 6 percent low education; senior ads required tertiary and mid-level education about equally often.

3.4 Automation exposure

Automation exposure comes from matching patents to the tasks of occupations, following Autor et al. (2024) and Oleš and Lábaj (2026). What the workers of an occupation do is described by the task statements of the 427 four-digit ISCO-08 occupations (Mihaylov and Tijdens, 2019). Each task statement is preceded by the prefix "a machine that automatically performs this work, replacing the human worker:", which gives the match a direction. Patents are represented by their title and abstract. Patents and task statements are embedded with PatentSBERTa, a Sentence-BERT model (Reimers and Gurevych, 2019) trained on patent texts. Let X_{pj} be the cosine similarity between patent p and task statement j. Because patents differ in how close they are to occupational language in general, each similarity is measured relative to the patent's median similarity, $\tilde{X}_{pj} = X_{pj} - \text{median}_{j'} X_{pj'}$, and a patent matches a task statement when $\tilde{X}_{pj}$ is in the top 15 percent of all patent–task pairs of the vintage. The number of matches of occupation o with a set of patents $\mathcal{P}$ is $M_o(\mathcal{P}) = \sum_{j \in o} \sum_{p \in \mathcal{P}} \mathbf{1}\{\tilde{X}_{pj} \geq \lambda\}$, where λ is the 85th percentile.

The sets of patents come from the AI labels of the USPTO AI Patent Dataset (Pairolero et al., 2024). A patent belongs to an AI type when the dataset predicts the AI type with a probability of at least 0.86, the threshold that balances precision and recall; "all AI" is the union of the eight AI types. I identify industrial robots in the same way, with patents in the robot classes of the Cooperative Patent Classification as positive examples. The main vintage consists of the 526,233 breakthrough patents granted between 1976 and 2015, identified with the indicator of Kelly et al. (2021); it includes 10,421 patents on natural language processing and 866 patents on industrial robots (Appendix Table B.3). As a check, I use all patents granted between 2010 and 2020.

Raw numbers of matches mostly measure how close the vocabulary of an occupation is to the vocabulary of patents in general. I therefore measure *excess exposure*, whether patents of

AI type f match occupation o more often than its matches to all patents predict,

$$x_{o,f} = \log_2 \frac{M_o(\mathcal{P}_f) + 1}{M_o(\mathcal{P})\, p_f + 1}, \qquad p_f = \frac{N_f}{N}, \tag{3}$$

where $\mathcal{P}$ is the set of all N patents of the vintage and $\mathcal{P}_f$ the set of the N_f patents of AI type f. Excess exposure is zero when patents of AI type f match the occupation in proportion to their share of all patents, and one when they match it twice as often. The main measure is automation exposure to natural language processing, the AI type closest to large language models. For the skill requirements, the automation exposure of a three-digit occupation is the average of its four-digit occupations, weighted by their pre-period job ads that state an experience requirement.

I group occupations into fifths of automation exposure, each with the same number of pre-period job ads that state an experience requirement. Table 1 lists the largest occupations of the *top fifth* and the *bottom fifth*. The top fifth consists of software developers, systems analysts, web developers, secretaries, inquiry clerks, personnel professionals and management analysts: occupations whose tasks consist of processing text, code and information. The bottom fifth consists of manufacturing laborers, freight handlers, truck drivers, cleaners, electrical mechanics and cooks, together with engineering professionals not elsewhere classified. Before 2022, the entry-level share was 56 percent in the top fifth and 49 percent in the bottom fifth.

Table 1: The largest occupations in the top fifth and in the bottom fifth of automation exposure

Occupation (ISCO-08)	Exposure (SD)	Ads stating experience (000)	Entry-level share (%)
Top fifth of automation exposure			
2512 Software Developers	0.95	2,686	50
2511 Systems Analysts	1.05	1,781	49
3412 Social Work Associate Professionals	1.00	1,079	81
3343 Administrative and Executive Secretaries	1.30	975	47
2423 Personnel and Careers Professionals	1.15	742	64
4225 Inquiry Clerks	1.00	664	73
2421 Management and Organization Analysts	1.74	611	42
2513 Web and Multimedia Developers	1.54	553	52
Bottom fifth of automation exposure			
2149 Engineering Professionals Not Elsewhere Classified	−1.49	1,801	53
9329 Manufacturing Labourers Not Elsewhere Classified	−1.28	1,547	54
9333 Freight Handlers	−1.23	1,065	50
8332 Heavy Truck and Lorry Drivers	−1.38	631	44
9112 Cleaners and Helpers in Offices, Hotels and Other Establishments	−2.38	613	46
7412 Electrical Mechanics and Fitters	−1.95	454	52
3123 Construction Supervisors	−1.37	440	29
5120 Cooks	−1.43	409	37

Notes: Four-digit ISCO-08 occupations with the most job ads that state an experience requirement in 2020–2022, summed over the cells of the sample. Exposure: automation exposure to natural language processing (Equation (3), breakthrough patents granted 1976–2015), in standard deviations across occupations weighted by pre-period job ads. Fifths contain equal numbers of pre-period job ads that state an experience requirement.

4 Empirical strategy

The design follows Brynjolfsson et al. (2025a), who regress the change in an occupation's employment after November 2022 on indicators for the occupation's fifth of AI exposure, separately for each age group. For cell c in country k, let $\Delta y_c = \bar{y}_c^{post} - \bar{y}_c^{pre}$ be the change in an outcome between the pre-period and the post-period. I estimate

$$\Delta y_c = \theta_k + \sum_{q=2}^{5} \beta_q \, \mathbf{1}\{q(c) = q\} + \varepsilon_c, \tag{4}$$

where $q(c)$ is the fifth of automation exposure of the cell's occupation and θ_k are country fixed effects. The coefficient β_5 compares the top fifth with the bottom fifth in the same country. The outcomes are the log number of entry-level ads and the log number of senior ads, estimated separately, as Brynjolfsson et al. (2025a) estimate young and older workers separately. I also report the coefficient of standardized automation exposure in place of the indicators, the change per standard deviation of exposure. The data are annual counts of job ads, so the effects over time are measured year by year: I replace Δy_c by the change between 2022 and each other year, $y_{ct} - y_{c,2022}$, which gives one coefficient per year relative to the last year before the release of ChatGPT. The average of the yearly coefficients in the post-period minus their average in the pre-period equals the pooled estimate.

For the skill requirements, the outcomes are the share of entry-level ads that require skill type g, the same share in senior ads, and their difference. The difference is the triple difference of Proposition 2: it compares entry-level ads with senior ads in the same cell, so anything that changes the skill requirements of all job ads of an occupation, such as a change in how its job ads are written, cancels out.

Cells are weighted by their pre-period job ads that state an experience requirement, so the estimates are effects for the average job ad, and the weights are fixed before the release of ChatGPT. Standard errors are clustered by occupation, four-digit for job ads and three-digit for skill requirements, the level at which exposure varies.

The design identifies the effect of generative AI if, without generative AI, outcomes in occupations with high and low automation exposure in the same country would have evolved in parallel. Three features support this assumption. First, exposure comes from patents granted by the USPTO at least seven years before the release of ChatGPT, which European job ads posted between 2020 and 2025 cannot have shaped. Second, exposure comes from texts and uses no outcome of European labor markets. Third, the release of ChatGPT made the capabilities of large language models available to every firm at the same time. Three tests address the remaining threats. The event study shows whether exposed occupations were on different paths before 2022. The robot placebo shows whether exposure to technology in general predicts changes after 2022. Finally, I exclude ICT occupations, whose hiring may have fallen after 2022 because of rising interest rates and the end of the pandemic hiring boom, and I restrict the job ads to the same sources, because the sources that the WIH collects changed over the period.

5 Results

5.1 Entry-level ads and senior ads

Table 2 reports the changes in the number of job ads. In the top fifth of automation exposure to natural language processing, the number of entry-level ads fell by 24 log points (standard error 0.08) after 2022 relative to the bottom fifth in the same country. Senior ads fell by 20 log points (0.07), and all job ads by 19 log points. Per standard deviation of automation exposure, entry-level ads fell by 9 log points and senior ads by 8 log points. Automation exposure to all AI gives almost the same estimates. Automation exposure to industrial robots predicts no change: the estimates are close to zero, with standard errors larger than the estimates.

Table 2: Job ads after ChatGPT, by automation exposure

	(1) Entry-level ads (log)	(2) Senior ads (log)	(3) All job ads (log)
Natural language processing			
Top fifth	−0.239***	−0.198***	−0.185***
	(0.082)	(0.067)	(0.066)
Per standard deviation	−0.093***	−0.083***	−0.069***
	(0.025)	(0.020)	(0.020)
All AI			
Top fifth	−0.249***	−0.237***	−0.197***
	(0.076)	(0.057)	(0.064)
Per standard deviation	−0.099***	−0.090***	−0.075***
	(0.021)	(0.017)	(0.019)
Industrial robots (placebo)			
Top fifth	−0.028	−0.010	−0.018
	(0.071)	(0.058)	(0.056)
Per standard deviation	−0.005	0.001	−0.000
	(0.022)	(0.017)	(0.017)
Cells		7,203	
Occupations (clusters)		411	

Notes: Coefficients from Equation (4): change in the log number of job ads between the pre-period (2020–22) and the post-period (2023–25). Top fifth: occupations in the top fifth of automation exposure relative to the bottom fifth, with indicators for the second to fourth fifths. Per standard deviation: coefficient of standardized automation exposure. All regressions include country fixed effects. Cells: country × four-digit occupation, weighted by pre-period job ads that state an experience requirement. Standard errors clustered by four-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent. Breakthrough patents granted 1976–2015.

Because entry-level ads and senior ads fell by similar amounts, the entry-level share did not change: in the top fifth it changed by −0.6 percentage points relative to the bottom fifth, with a standard error of 0.6. By Proposition 1, generative AI automated entry-level tasks and senior tasks roughly in proportion. This is the main difference from the United States, where Brynjolfsson et al. (2025a) find that employment fell for young workers in exposed occupations but not for older workers.

Figure 1 shows the changes over time. Before 2022, entry-level ads in the top fifth evolved like those in the bottom fifth: relative to 2022, the coefficients are 0.05 in 2020 and −0.07 in 2021. After 2022, entry-level ads in the top fifth fell by 15 log points in 2023 and by 30 and 28 log points in 2024 and 2025. Senior ads in the top fifth grew faster than in the bottom fifth before 2022, by 11 to 16 log points, and then fell by 23 log points in 2023, 29 in 2024 and 36 in 2025. For industrial robots, neither entry-level ads nor senior ads changed.

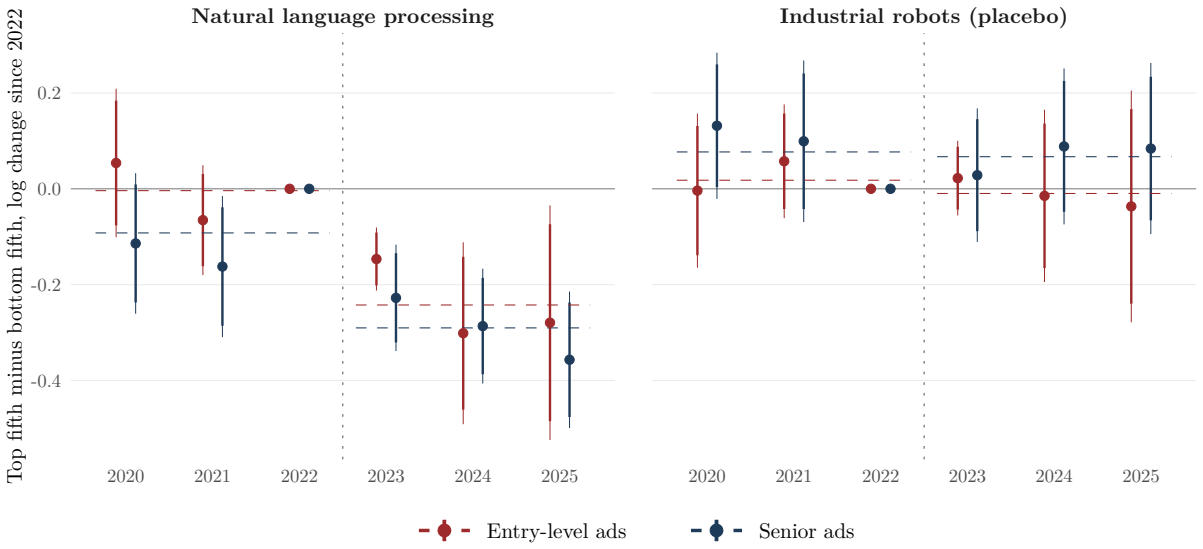


Figure 1: Entry-level ads and senior ads around the release of ChatGPT: top fifth relative to bottom fifth of automation exposure.

Notes: Coefficients β_5 of Equation (4) with the change between 2022 and each year, $y_{ct} - y_{c,2022}$, as the outcome: log number of entry-level ads and of senior ads in the top fifth of automation exposure relative to the bottom fifth, within country. The data are annual: each point is the coefficient for one year. Dashed lines are the averages of the yearly coefficients in the pre-period (2020–22, including the base year 2022) and in the post-period (2023–25); the distance between them equals the pooled estimate in Table 2. Thick bars are 90 percent and thin bars 95 percent confidence intervals, clustered by four-digit occupation. Breakthrough patents granted 1976–2015.

5.2 Skill requirements

Generative AI changed what entry-level jobs require (Table 3 and Figure 2). Per standard deviation of automation exposure to natural language processing, the number of skill requirements per entry-level ad fell by 4.7 percent. The share of entry-level ads that require digital skills fell by 1.4 percentage points, analytical skills by 1.3, language skills by 1.1, and social, interpersonal and people-management skills by 1.0 each. Manual skill requirements did not change, and neither did writing skill requirements, which only 2 percent of entry-level ads mention. In the top fifth relative to the bottom fifth, the declines are more than twice as large: 2.8 percentage points for digital, 3.1 for analytical, and 2.5 for language skill requirements. Senior ads lost the same skill types. By Proposition 2, generative AI automated the tasks that require language, digital, analytical, social, interpersonal and people-management skills more than proportionally, in entry-level jobs and in senior jobs.

The robot placebo shows that the declines are specific to generative AI. Automation exposure to industrial robots does not lower any skill requirement. It raises the share of job ads that require language skills, by 0.9 percentage points in entry-level ads and 0.7 in senior ads, the opposite of what automation exposure to natural language processing does.

The declines grew over time (Figure 3). In 2023, the first year after the release of ChatGPT, skill requirements in entry-level ads changed little, except analytical skill requirements, which fell by 0.6 percentage points. In 2024, the shares of entry-level ads that require digital, language, analytical, interpersonal and people-management skills fell by 1.2 to 1.5 percentage points per

Table 3: Skill requirements after ChatGPT, per standard deviation of automation exposure

	Natural language processing			Industrial robots (placebo)	
	(1) Entry-level ads	(2) Senior ads	(3) Difference	(4) Entry-level ads	(5) Senior ads
Skill requirements per ad (log)	-0.047^{***}	-0.054^{***}	0.007	0.016	0.012
	(0.012)	(0.012)	(0.007)	(0.011)	(0.013)
Share of ads that require the skill type (percentage points)					
Language	-1.10^{***}	-1.32^{***}	0.21	0.95^{***}	0.69^{***}
	(0.30)	(0.24)	(0.21)	(0.29)	(0.20)
Digital	-1.41^{***}	-1.78^{***}	0.37	0.36	0.71^{**}
	(0.33)	(0.28)	(0.30)	(0.27)	(0.35)
Analytical	-1.34^{***}	-1.03^{***}	−0.31	0.09	−0.07
	(0.37)	(0.33)	(0.19)	(0.40)	(0.38)
Social	-1.01^{***}	-0.75^{***}	−0.26	0.15	0.00
	(0.28)	(0.24)	(0.21)	(0.28)	(0.27)
Interpersonal	-1.02^{**}	-0.99^{**}	−0.03	0.09	−0.11
	(0.44)	(0.43)	(0.19)	(0.42)	(0.36)
People management	-1.04^{***}	-0.77^{***}	-0.27^{*}	0.23	0.20
	(0.22)	(0.17)	(0.15)	(0.27)	(0.19)
Writing	−0.01	−0.05	0.04	−0.07	−0.14
	(0.09)	(0.13)	(0.08)	(0.07)	(0.12)
Manual	−0.13	0.13	−0.26	−0.10	−0.31
	(0.30)	(0.27)	(0.16)	(0.20)	(0.19)
Cells			2,516		
Occupations (clusters)			122		

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of automation exposure, with country fixed effects. Columns (1) and (4): entry-level ads; columns (2) and (5): senior ads; column (3): entry-level ads minus senior ads in the same cell. First row: log number of skill requirements per job ad. Other rows: share of job ads that require at least one skill of the skill type, in percentage points. Cells: country × three-digit occupation, weighted by pre-period job ads that state an experience requirement; automation exposure of a three-digit occupation is the weighted average of its four-digit occupations. Standard errors clustered by three-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

standard deviation of exposure relative to 2022, and in 2025 the declines reached 2.1 percentage points for digital skills, 1.9 for language skills, 1.4 to 1.5 for analytical, interpersonal and people-management skills, and 1.0 for social skills. The number of skill requirements per entry-level ad was 8 percent lower per standard deviation in 2025. The timing matches the adoption of generative AI by firms, which spread through 2024 and 2025. Before 2022, the coefficients are small in 2021, but several are positive in 2020, the first year of the pandemic, by 0.4 to 0.8 percentage points: skill requirements in exposed occupations fell somewhat between 2020 and 2021 as well, and the post-period estimates should be read against this movement.

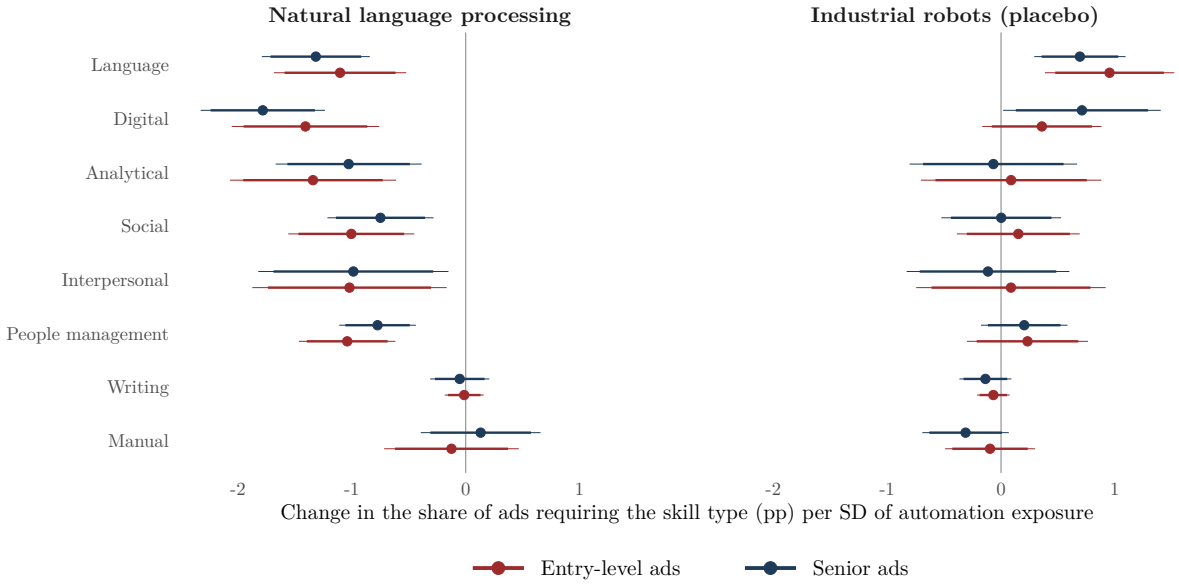


Figure 2: Skill requirements of entry-level ads and senior ads after ChatGPT, by skill type.
Notes: Change between the pre-period and the post-period in the share of job ads that require at least one skill of the skill type, per standard deviation of automation exposure, with country fixed effects; Table 3. Thick bars are 90 percent and thin bars 95 percent confidence intervals, clustered by three-digit occupation.

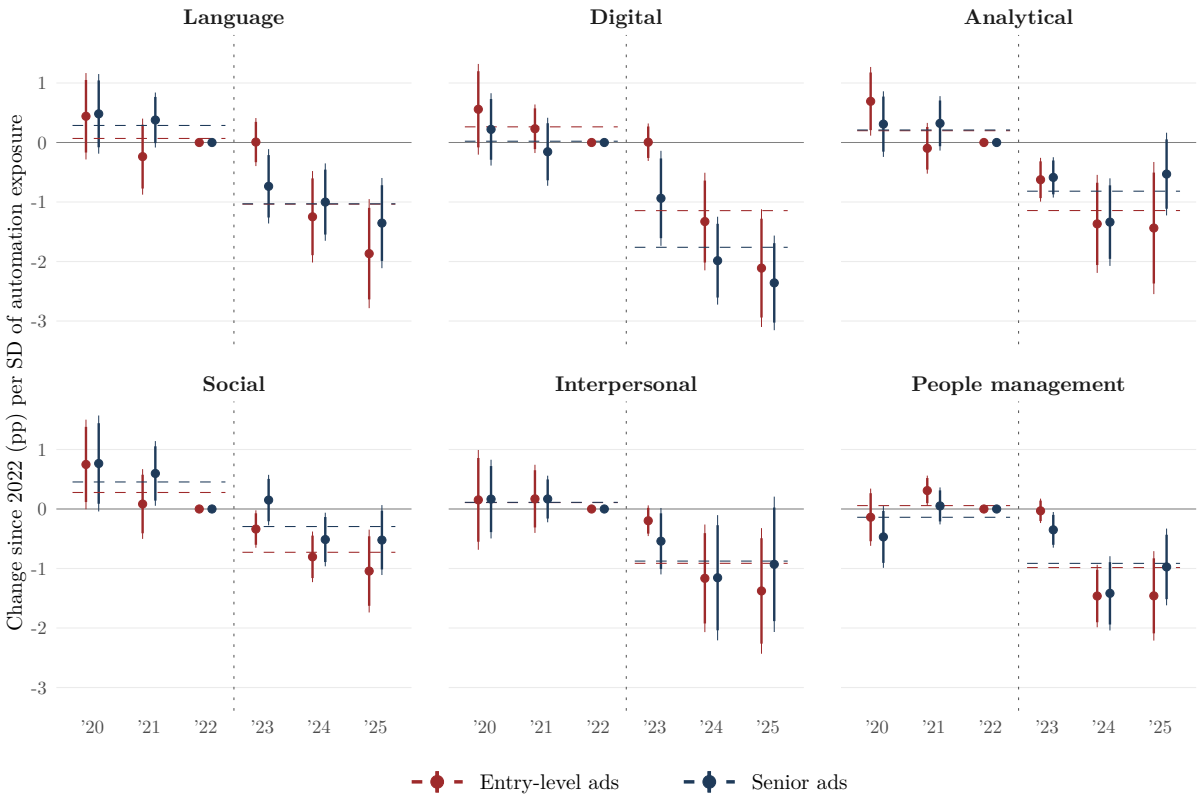


Figure 3: Skill requirements of entry-level ads and senior ads over time.

Notes: Change between 2022 and each year in the share of job ads that require at least one skill of the skill type, per standard deviation of automation exposure to natural language processing, with country fixed effects. The data are annual: each point is the coefficient for one year. Dashed lines are the averages of the yearly coefficients in the pre-period (including the base year 2022) and in the post-period; the distance between them equals the pooled estimate in Table 3. Thick bars are 90 percent and thin bars 95 percent confidence intervals, clustered by three-digit occupation.

5.3 Entry-level ads compared with senior ads

Two skill types fell more in entry-level ads than in senior ads. Per standard deviation of automation exposure, analytical skill requirements fell by 1.34 percentage points in entry-level ads and by 1.03 in senior ads, and people-management skill requirements by 1.04 and 0.77 (Table 3, column 3). The differences, −0.31 and −0.27 percentage points, are about a quarter of the declines in entry-level ads. They grew over time: in 2025, the difference for analytical skill requirements reached −0.91 percentage points (standard error 0.34) and the difference for people-management skill requirements −0.49 (0.24) (Appendix Figure B.1). With automation exposure to all AI, the difference for people-management skill requirements is −0.40 (0.12). Language and digital skill requirements, by contrast, fell slightly, but not significantly, more in senior ads.

By Proposition 2, these differences suggest that generative AI automates the analytical and coordinating tasks of entry-level jobs more than proportionally. The pooled differences are significant only at the 10 percent level or just above it, so the evidence is weaker than for the declines themselves. Entry-level jobs in exposed occupations used to include analyzing information, solving problems and organizing the work of others; after 2022, firms ask less often for these skills in entry-level ads than in senior ads. The first rung remains, but it requires less of the work through which workers used to learn the analytical and coordinating tasks of senior jobs.

5.4 Skill requirements by education level

Exposed occupations also changed the education that their job ads require. Per standard deviation of automation exposure, the share of senior ads that require tertiary education fell by 0.59 percentage points and the share that requires mid-level education rose by 0.68 percentage points; in entry-level ads, the share that requires mid-level education rose and the share that requires low education fell (Appendix C). Does this shift explain the decline in skill requirements? If job ads that require mid-level education list fewer skill requirements than job ads that require tertiary education, a shift from tertiary to mid-level education lowers skill requirements even when the skill requirements at each education level do not change. I therefore add to the regressions of Table 3 the changes in the shares of entry-level ads and of senior ads in the cell that require tertiary education and low education. The skill requirements of entry-level ads hardly move: per standard deviation of automation exposure, the share of entry-level ads that require analytical skills falls by 1.32 percentage points instead of 1.34, and the shares that require digital and language skills fall by 1.41 and 1.10 percentage points, as without the controls. In senior ads, the decline of analytical skill requirements shrinks from 1.03 to 0.88 percentage points. The difference between entry-level ads and senior ads in analytical skill requirements therefore grows slightly, from −0.31 to −0.35 percentage points, whereas the difference in people-management skill requirements shrinks from −0.27 to −0.19 percentage points and is no longer significant. Because automation exposure also changed the education requirements, these estimates do not isolate a causal channel; they show that the shift in education requirements accounts for little of the decline in skill requirements.

A more direct test compares entry-level ads and senior ads that require the same education level. For this test, I split each skill cell by the education level that its job ads require, so a

cell is a combination of country, three-digit occupation and education level, and I estimate the regressions of Table 3 separately for each education level (Figure 4; Appendix Table C.2 reports all estimates). Entry-level ads lost skill requirements at every education level: per standard deviation of automation exposure, the number of skill requirements per entry-level ad fell by 4.6 percent in job ads that require tertiary education, by 4.2 percent in job ads that require mid-level education, and by 7.0 percent in job ads that require low education. The education levels differ in which skill requirements entry-level ads lost relative to senior ads. In job ads that require mid-level education, entry-level ads and senior ads lost the same skill types by similar amounts: digital skill requirements fell by 1.99 and 2.09 percentage points, analytical skill requirements by 1.67 and 1.31, and people-management skill requirements by 1.31 and 1.16. In job ads that require tertiary education, entry-level ads lost social, people-management and analytical skill requirements, by 0.89, 0.62 and 0.90 percentage points, whereas senior ads lost much less of them, 0.21, 0.20 and 0.55 percentage points, and lost more digital skill requirements instead, 1.52 against 0.84. The differences between entry-level ads and senior ads that require tertiary education, −0.67, −0.42 and −0.36 percentage points for social, people-management and analytical skill requirements, are imprecise, with standard errors of 0.30 to 0.43. Entry-level ads that require low education, which account for 5 percent of the pre-period job ads of these cells, lost language, digital and social skill requirements. Pooled over education levels with country × education fixed effects, the shares of entry-level ads that require digital, analytical and language skills fall by 1.43, 1.26 and 1.18 percentage points, almost as much as in Table 3, and the difference between entry-level ads and senior ads in analytical skill requirements is −0.32 percentage points (standard error 0.20).

The number of job ads by education level adds little (Figure 5). Entry-level ads fell at every education level, and senior ads in job ads that require tertiary and mid-level education. Only in job ads that require mid-level education did entry-level ads fall relative to senior ads, by 3.0 log points per standard deviation, and the yearly estimates show that this relative decline began before 2022. The first rung therefore lost skill requirements at every education level; in jobs that require tertiary education, it lost more social, people-management and analytical skill requirements than senior jobs, although this difference is imprecise.

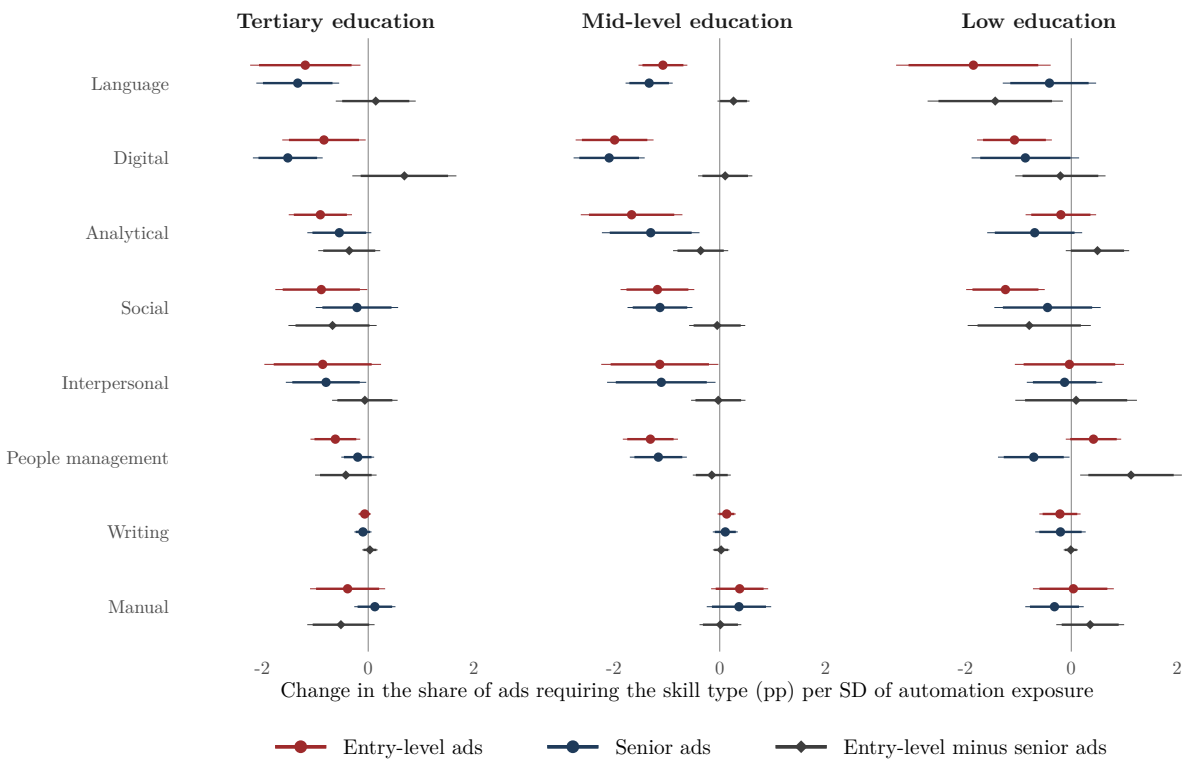


Figure 4: Skill requirements of entry-level ads and senior ads by education level.

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) in the share of job ads that require at least one skill of the skill type, per standard deviation of automation exposure to natural language processing, with country fixed effects, estimated separately for job ads that require tertiary, mid-level and low education. Cells: country × three-digit occupation × education level, with at least ten entry-level ads and ten senior ads in every year, weighted by their pre-period entry-level ads and senior ads. Diamonds: entry-level ads minus senior ads in the same cell. Thick bars are 90 percent and thin bars 95 percent confidence intervals, clustered by three-digit occupation. Estimates in Appendix Table C.2.

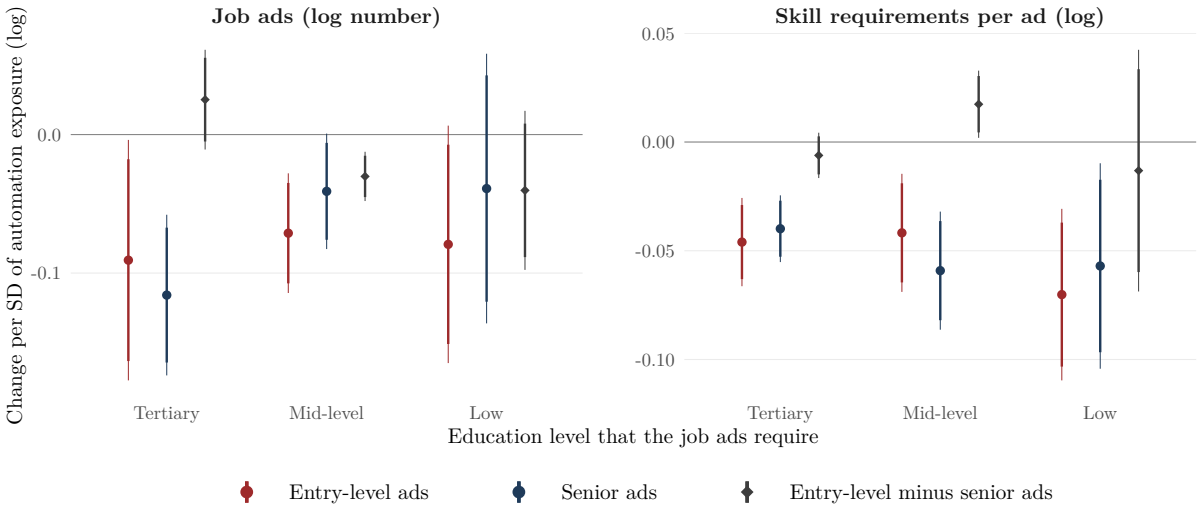


Figure 5: Job ads and skill requirements per job ad by education level.

Notes: Change between the pre-period and the post-period in the log number of job ads and in the log number of skill requirements per job ad, per standard deviation of automation exposure to natural language processing, with country fixed effects, estimated separately for each education level that the job ads require. Cells, diamonds and confidence intervals as in Figure 4. Estimates in Appendix Table C.2.

5.5 Robustness

Table 4 repeats all main estimates under alternative specifications, and every column reports the same outcomes. The decline in skill requirements holds in every specification. In entry-level ads, the number of skill requirements per job ad and the shares that require language, digital, analytical, social, interpersonal and people-management skills fall in all nine columns, while writing and manual skill requirements change little and never significantly at the 5 percent level. The declines in language, digital and analytical skill requirements are between 0.46 and 1.85 percentage points per standard deviation and significant at the 5 percent level in every column, except digital skill requirements within source, which are significant at the 10 percent level. The declines are smaller among job ads from continuous sources, the sources that the WIH observes in every quarter between 2020 and 2025, and within source, when each cell is split by source so that only job ads from the same source are compared. Among job ads from continuous sources, the decline of social skill requirements is no longer significant, and within source, the decline of people-management skill requirements is no longer significant.

The differences between entry-level ads and senior ads are less robust. The difference in people-management skill requirements is negative in every column, between −0.21 and −0.58 percentage points, and significant at the 10 percent level or better in six of the nine. The difference in analytical skill requirements is negative in eight columns, between −0.26 and −0.53 percentage points, and largest without the United Kingdom and within source, but it is positive, at 0.26 percentage points and not significant, with equal country weights, which give small countries the same weight as large ones. Language and digital skill requirements fell less in entry-level ads than in senior ads in every column, mostly not significantly.

The decline in the number of job ads holds within industries, without the United Kingdom, without Poland, without ICT occupations, with equal country weights, and with exposure from patents granted between 2010 and 2020, with estimates between 5 and 12 log points per standard deviation. Excluding ICT occupations matters: the estimates fall from 9 to 7 log points for entry-level ads, so part of the decline reflects the fall in hiring of software developers and other ICT occupations after 2022. Among job ads from continuous sources and within source, however, the number of job ads does not change with automation exposure. The decline in the number of job ads therefore comes from sources that the WIH does not observe throughout the period, and it may reflect changes in these sources rather than in the demand of firms. The changes in skill requirements are smaller within the same sources, but they remain.

Two further checks are reported in Appendix C. First, senior ads include job ads that require one or two years of experience, which may differ little from entry-level ads. With senior ads defined as job ads that require four years of experience or more, the skill requirements of entry-level ads and senior ads fall by similar amounts as in Table 3, and the differences between them are no larger (Appendix Table C.4); the number of entry-level ads falls relative to the number of job ads that require four years or more, but this relative decline began before 2022 (Appendix Table C.3). Second, the patents that automate tasks of an occupation can also create new tasks in it. *Augmentation exposure*, which matches the same patents to what an occupation produces rather than to what its workers do (Autor et al., 2024), is correlated with automation exposure across three-digit occupations at 0.82. Holding augmentation exposure

Table 4: Robustness: per standard deviation of automation exposure to natural language processing

	(1) Main	(2) Country × industry FE	(3) Without United Kingdom	(4) Without Poland	(5) Without ICT occupations	(6) Equal country weights	(7) Patents 2010–20	(8) Continuous sources	(9) Within source
Job ads (log number)									
Entry-level ads	−0.093***	−0.062***	−0.109***	−0.089***	−0.068***	−0.090***	−0.091***	−0.032	−0.018
	(0.025)	(0.020)	(0.029)	(0.025)	(0.023)	(0.019)	(0.025)	(0.037)	(0.030)
Senior ads	−0.083***	−0.054***	−0.122***	−0.077***	−0.057***	−0.077***	−0.083***	−0.032	−0.008
	(0.020)	(0.017)	(0.022)	(0.019)	(0.016)	(0.021)	(0.019)	(0.050)	(0.037)
Entry-level ads									
Skill requirements	−0.047***	−0.040***	−0.059***	−0.047***	−0.042***	−0.031***	−0.047***	−0.046***	−0.033***
per ad (log)	(0.012)	(0.009)	(0.010)	(0.013)	(0.012)	(0.009)	(0.011)	(0.010)	(0.010)
Language	−1.10***	−1.19***	−1.71***	−0.92***	−1.27***	−0.46**	−1.17***	−1.48***	−1.85***
	(0.30)	(0.30)	(0.37)	(0.29)	(0.28)	(0.21)	(0.30)	(0.40)	(0.47)
Digital	−1.41***	−1.40***	−1.36***	−1.28***	−1.39***	−0.93***	−1.37***	−0.88**	−0.55*
	(0.33)	(0.25)	(0.31)	(0.34)	(0.36)	(0.25)	(0.32)	(0.35)	(0.33)
Analytical	−1.34***	−0.84***	−1.32***	−1.22***	−1.10***	−0.72***	−1.34***	−0.80**	−0.81**
	(0.37)	(0.25)	(0.30)	(0.37)	(0.35)	(0.26)	(0.36)	(0.34)	(0.34)
Social	−1.01***	−0.79***	−1.39***	−0.98***	−0.98***	−0.81***	−1.01***	−0.51	−0.42*
	(0.28)	(0.21)	(0.30)	(0.30)	(0.30)	(0.30)	(0.28)	(0.37)	(0.25)
Interpersonal	−1.02**	−0.78**	−1.14***	−0.88*	−0.75*	−1.01***	−1.05**	−0.75**	−0.77**
	(0.44)	(0.34)	(0.38)	(0.48)	(0.43)	(0.27)	(0.42)	(0.37)	(0.32)
People management	−1.04***	−0.70***	−1.01***	−0.98***	−1.02***	−1.49***	−1.02***	−0.43*	−0.33
	(0.22)	(0.22)	(0.23)	(0.21)	(0.25)	(0.21)	(0.21)	(0.23)	(0.21)
Writing	−0.01	0.03	−0.07*	0.00	0.05	−0.14*	−0.00	0.07	0.00
	(0.09)	(0.08)	(0.04)	(0.09)	(0.07)	(0.08)	(0.08)	(0.06)	(0.07)
Manual	−0.13	−0.17	−0.19	−0.24	0.04	0.05	−0.14	0.05	0.37
	(0.30)	(0.24)	(0.31)	(0.31)	(0.30)	(0.33)	(0.28)	(0.24)	(0.29)
Entry-level ads minus senior ads in the same cell									
Skill requirements	0.007	0.001	0.004	0.002	0.003	0.018***	0.007	0.011	−0.009
per ad (log)	(0.007)	(0.006)	(0.008)	(0.005)	(0.006)	(0.006)	(0.006)	(0.007)	(0.008)
Language	0.21	0.13	0.04	0.23	0.14	0.48**	0.18	0.74***	0.31
	(0.21)	(0.15)	(0.29)	(0.20)	(0.22)	(0.19)	(0.22)	(0.25)	(0.21)
Digital	0.37	0.23	0.28	0.47	0.37	0.38**	0.40	0.11	0.23
	(0.30)	(0.26)	(0.39)	(0.32)	(0.33)	(0.16)	(0.29)	(0.25)	(0.25)
Analytical	−0.31	−0.37**	−0.49**	−0.26	−0.30	0.26	−0.36*	−0.43*	−0.53**
	(0.19)	(0.19)	(0.23)	(0.20)	(0.21)	(0.20)	(0.19)	(0.23)	(0.27)
Social	−0.26	−0.26	−0.34	−0.22	−0.23	0.34*	−0.32	0.23	−0.36
	(0.21)	(0.19)	(0.25)	(0.22)	(0.22)	(0.18)	(0.20)	(0.23)	(0.25)
Interpersonal	−0.03	−0.14	−0.09	−0.06	−0.04	−0.26	−0.08	0.29	−0.17
	(0.19)	(0.16)	(0.20)	(0.21)	(0.21)	(0.18)	(0.19)	(0.18)	(0.21)
People management	−0.27*	−0.31*	−0.53***	−0.34*	−0.21	−0.58***	−0.27*	−0.25	−0.29
	(0.15)	(0.18)	(0.18)	(0.20)	(0.16)	(0.12)	(0.15)	(0.21)	(0.24)
Writing	0.04	0.00	−0.05*	0.05	−0.04	0.05	0.04	−0.03	0.00
	(0.08)	(0.06)	(0.02)	(0.08)	(0.05)	(0.05)	(0.07)	(0.03)	(0.03)
Manual	−0.26	−0.17	−0.08	−0.35**	−0.32*	0.18	−0.26*	0.13	0.16
	(0.16)	(0.13)	(0.24)	(0.17)	(0.17)	(0.14)	(0.15)	(0.20)	(0.17)
Cells									
Job ads	7,203	30,364	6,823	6,883	6,819	7,203	7,203	5,458	10,946
Skill requirements	2,516	11,580	2,400	2,408	2,393	2,516	2,516	1,883	3,555

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of automation exposure to natural language processing; every column reports the same outcomes. Job ads: log number of entry-level ads and of senior ads; cells of country × four-digit occupation, standard errors clustered by four-digit occupation. Skill requirements: log number of skill requirements per job ad and share of job ads that require at least one skill of the skill type (percentage points), for entry-level ads and for entry-level ads minus senior ads in the same cell; cells of country × three-digit occupation, standard errors clustered by three-digit occupation. Column (2): cells split by industry section, with country × industry fixed effects. Column (5): without ISCO-08 groups 25, 35 and 133. Column (6): weights rescaled so that every country has the same total weight. Column (7): exposure from all patents granted 2010–2020. Column (8): job ads from sources with at least 20 job ads in the country in every quarter 2020–2025. Column (9): cells split by continuous source, with country × source fixed effects. Standard errors in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

fixed, automation exposure lowers the skill requirements of entry-level ads more than in Table 3, digital skill requirements, for example, by 2.67 instead of 1.41 percentage points per standard deviation, because augmentation exposure raises them (Appendix Table C.5). The automation estimates for skill requirements are therefore, if anything, lower bounds; the estimates for the number of job ads do not change.

5.6 Other AI types

Natural language processing is one of eight AI types in the USPTO AI Patent Dataset. Appendix Table B.4 repeats the main estimates with automation exposure to each of them. Automation exposure to machine learning, evolutionary computation, speech, planning and control, knowledge processing and AI hardware gives estimates close to those for natural language processing: entry-level ads fall by 6 to 10 log points and senior ads by 5 to 9 log points per standard deviation, and the shares of entry-level ads that require language, digital, analytical, social, interpersonal and people-management skills fall by 0.7 to 1.6 percentage points, except interpersonal skill requirements with exposure to speech (0.46 percentage points, not significant). The estimates for speech are the smallest of these. Unlike natural language processing, several of these AI types also predict a decline in manual skill requirements of entry-level ads, by 0.4 to 0.6 percentage points, significant at the 5 percent level for evolutionary computation and AI hardware, and for machine learning and evolutionary computation, manual skill requirements fell significantly more in entry-level ads than in senior ads. The difference between entry-level ads and senior ads in people-management skill requirements is between −0.34 and −0.41 percentage points and significant at the 1 percent level for all AI and for every one of these AI types except speech; the difference in analytical skill requirements is negative for all of them but significant only for all AI, at the 10 percent level. Across four-digit occupations, automation exposure to these AI types is highly correlated with automation exposure to natural language processing, at 0.79 to 0.89, except for evolutionary computation at 0.63, so the data cannot tell them apart. They all reach occupations whose tasks consist of processing text, code and information.

Computer vision is the exception. Its automation exposure correlates with that to natural language processing at only 0.30, because it reaches visual professions and some manual occupations. Automation exposure to computer vision predicts no change in entry-level ads, senior ads or skill requirements, apart from a decline in writing skill requirements of entry-level ads that is significant only at the 10 percent level. The effects therefore come from AI that processes text, code and information, not from exposure to AI in general.

6 Discussion and conclusion

Generative AI has not removed the first rung of the career ladder in Europe. In occupations with high automation exposure to natural language processing, firms posted fewer entry-level ads after 2022, but they also posted fewer senior ads, and the entry-level share did not change. In the United States, Brynjolfsson et al. (2025a) find that employment fell for workers aged 22 to 25 in exposed occupations but not for older workers. The two results measure different things, employment by age in one case and posted demand by experience requirement in the other. In

European job ads, the decline in exposed occupations is not specific to entry-level jobs.

Generative AI has instead changed what the first rung requires. In exposed occupations, entry-level ads ask less often for language, digital, analytical, social, interpersonal and people-management skills, the skill types that large language models now supply, and the change grew with the adoption of generative AI in 2024 and 2025. Analytical and people-management skill requirements fell more in entry-level ads than in senior ads. The decline of analytical skill requirements matches the decline in the demand for skills required for non-routine analytical tasks in German establishments with AI activity (Peede and Stops, 2026). People-management skills, by contrast, went with lower automation exposure in Slovak job ads in 2022 (Oleš, 2026); that exposed occupations now ask for them less often, and less often in entry-level ads than in senior ads, suggests that generative AI reaches a skill type that earlier automation technologies left to workers. If workers learn the analytical and coordinating tasks of senior jobs by performing simpler versions of them in entry-level jobs, entry-level jobs that require less of these tasks teach less of them. The field experiment of Autor et al. (2026) points in the same direction: less experienced patent lawyers who worked with AI assistance did not improve at a core task once they worked without it. Studies that count entry-level jobs miss this change.

The conclusions have limits. Job ads measure posted demand, not hires, and an experience requirement or a skill requirement is a signal, not a binding rule. The WIH extracts skills from the texts of job ads, so a change in how job ads are written could change the measured skill requirements; the comparison of entry-level ads with senior ads in the same cell removes changes that affect all job ads of an occupation, and the decline holds within the same sources. The decline in the number of job ads depends on the sources and should be read with caution. Automation exposure measures what AI can do in an occupation, not whether firms use AI. Finally, the data cover the first three years after the release of ChatGPT.

For policy, the results suggest monitoring not only how many entry-level jobs remain open, but also what these jobs require. A first rung that asks less of the analytical and coordinating work of senior jobs may leave the career ladder in place while removing the work through which workers learn to climb it.

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Declaration of generative AI use

This paper is my independent work. I formulated the research question, selected the data, developed the exposure measures based on my prior research, and conducted all analyses within the secure WIH environment. I used Claude (Anthropic; Claude Opus 5.5) under my supervision to assist with coding, drafting, and editing, using only aggregated data. I reviewed all AI-generated outputs and take full responsibility for the paper's methodology, findings, and content.

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A Descriptive statistics

This appendix documents the job ads behind the design choices of the paper: which countries and sources the WIH covers, how it codes the experience requirement, and how the two countries whose data changed most, the United Kingdom and Poland, affect the estimates.

Table A.1 describes the job ads by country. The WIH collected 250 million job ads in the 27 member states of the European Union and the United Kingdom between 2020 and 2025, and 53 percent of them state an experience requirement. The share ranges from 23 percent in Germany to 82 percent in Finland, and the entry-level share from 15 percent in Lithuania to 84 percent in Estonia, which is why every comparison in the paper is made within countries. The education requirement is stated in 98 percent of job ads. The share of job ads that come from continuous sources, the 170 sources that the WIH observes in every quarter between 2020 and 2025, rose from 43 percent in the pre-period to 54 percent in the post-period, because other sources left the data. The cells of the sample cover 98 percent of the job ads that state an experience requirement.

Table A.2 shows how the experience requirement is coded. An entry-level ad is a job ad in the bands "no experience" and "less than one year". Between 2020 and 2025, job ads moved from the band "less than one year" to the band "no experience", from 45 and 7 percent of job ads that state an experience requirement to 22 and 26 percent. The move also occurs without the United Kingdom, where most job ads are coded "less than one year", so it reflects how job ads state short experience requirements or how the WIH codes them. Because both bands count as entry-level ads, the move does not change the entry-level share: without the United Kingdom, the entry-level share was 44 percent in both 2022 and 2025. This is one reason to define entry-level ads by both bands rather than by either one.

Table A.3 shows the two countries whose data changed most. In the United Kingdom, the number of job ads fell from 12.0 million in 2020 and 10.5 million in 2022 to 3.0 million in 2025, and the United Kingdom's share of job ads that state an experience requirement fell from 41 to 10 percent. A single job portal, source 638, contributed 3.5 million job ads a year in 2021 and 2022 and none from 2024. Within the United Kingdom, the share of job ads that state an experience requirement fell from about 70 to 62 percent in 2024, and the entry-level share stayed between 73 and 77 percent. Because the entry-level share is higher in the United Kingdom than elsewhere, the fall in its job ads lowers the entry-level share of all job ads (Table B.1) but not the comparisons within countries. In Poland, the coding of the experience requirement changed in 2021: the share of job ads that state an experience requirement rose from 30 to 69 percent, and the share coded "no experience" from 10 to 64 percent, so the entry-level share rose from 36 to 75 percent. The change affects all Polish job ads and falls in the pre-period. The country fixed effects absorb the part of it that is common to all Polish occupations, and the estimates without the United Kingdom and without Poland in Table 4 are similar to the main estimates; without the United Kingdom, the declines in language and social skill requirements and the differences in analytical and people-management skill requirements are larger.

Table A.4 shows how the sample and the main estimates depend on the minimum number of job ads per cell and year. With a minimum of 5, 10, 20 or 50 job ads, the cells cover between 98 and 99 percent of the job ads that state an experience requirement and almost all entry-level

ads and senior ads of the skill sample, and the estimates of skill requirements hardly change, because cells are weighted by their job ads. The decline in the number of job ads is somewhat smaller with a minimum of five, 7 instead of 9 log points per standard deviation for entry-level ads. I use a minimum of ten.

Table A.1: Job ads by country

	Job ads (million)	Stating exp. (%)	Entry-level share (%)	Stating educ. (%)	Contin. sources	Ads from contin. sources (%)		Cells	Ads in cells (%)
	2020–25	2020–25	2020–22	2020–25		2020–22	2023–25		2020–25
AT	3.1	31	34	100	7	46	68	244	98
BE	9.2	64	41	100	14	51	65	366	98
BG	0.9	43	65	99	4	13	28	221	94
CY	0.2	47	53	100	1	3	1	109	83
CZ	2.8	53	57	100	8	62	53	251	97
DE	47.4	23	34	100	19	63	75	363	99
DK	1.5	34	54	3	1	36	23	223	97
EE	0.4	61	84	100	2	26	29	175	94
EL	0.5	39	53	100	3	59	69	182	94
ES	5.2	75	27	100	8	67	65	341	96
FI	0.9	82	41	100	1	14	18	268	96
FR	72.2	63	35	100	15	36	51	380	99
HR	0.9	61	30	100	4	57	83	238	92
HU	1.4	42	64	45	4	46	53	239	97
IE	2.2	69	67	100	5	7	19	246	97
IT	14.8	42	32	100	11	27	51	351	99
LT	0.9	68	15	100	4	58	46	261	98
LU	0.2	60	35	100	1	5	2	120	90
LV	0.6	27	45	100	4	47	52	172	93
MT	0.1	62	70	100	4	65	80	86	84
NL	12.8	50	62	100	7	8	12	346	98
PL	10.8	57	73	100	6	82	69	320	97
PT	2.2	52	52	31	5	34	34	291	99
RO	1.8	47	55	100	8	73	68	264	98
SE	7.1	39	32	100	2	54	80	350	100
SI	0.5	46	22	99	3	54	55	176	93
SK	1.3	52	61	100	3	52	82	240	96
UK	48.1	68	76	100	16	32	43	380	99
All	249.7	53	53	98	170	43	54	7,203	98

Notes: All job ads of the WIH, 2020–2025, assigned to the first country of the job ad. Stating exp. and stating educ.: share of job ads that state an experience requirement and an education requirement. Entry-level share: entry-level ads as a share of job ads that state an experience requirement. Contin. sources: number of sources with at least 20 job ads in the country in every quarter 2020–2025; ads from contin. sources: share of job ads that come from at least one of them. Cells: country × four-digit occupation cells of the sample. Ads in cells: share of the job ads that state an experience requirement that belong to the cells of the sample.

Table A.2: The experience requirement by year

	Stating	Share of job ads stating experience (%)						
	experience (%)	None	<1 year	1–2	2–4	4–6	>6	Other
2020	48.6	7.2	44.5	11.6	16.6	4.1	16.1	0.0
2021	53.1	13.1	41.2	11.9	16.1	3.7	14.0	0.0
2022	52.8	15.7	38.1	12.6	14.4	3.0	16.2	0.0
2023	51.1	14.4	35.7	16.0	13.8	3.3	16.8	0.0
2024	56.7	25.3	22.9	16.0	15.9	3.1	16.8	0.0
2025	56.1	25.6	21.5	14.3	17.6	3.6	17.4	0.0

Notes: All job ads of the WIH in the 27 member states of the European Union and the United Kingdom. Stating experience: share of all job ads. Other columns: experience bands as coded by the WIH, as a share of job ads that state an experience requirement. Entry-level ads: bands "None" and "<1 year".

Table A.3: The United Kingdom and Poland

	United Kingdom					Poland			
	Job ads (million)	Source 638 (mill.)	Share of all ads (%)	Stating exp. (%)	Entry-level share (%)	Job ads (million)	Stating exp. (%)	No experience (%)	Entry-level share (%)
2020	12.0	2.2	40.8	68	73	1.45	30	10	36
2021	9.8	3.5	32.2	72	77	1.40	69	64	75
2022	10.5	3.5	30.1	69	77	2.41	74	70	80
2023	8.6	2.8	23.5	68	77	2.39	43	46	63
2024	4.2	0.0	11.7	61	73	1.84	60	54	65
2025	3.0	0.0	9.9	62	75	1.28	63	63	72

Notes: Source 638: the largest job portal of the United Kingdom in the WIH, flagged as stable by the WIH; job ads by year. Share of all ads: the United Kingdom's share of the job ads that state an experience requirement in the 27 member states of the European Union and the United Kingdom. Stating exp.: share of job ads that state an experience requirement. No experience: share of job ads coded "no experience" among job ads that state an experience requirement.

Table A.4: Minimum number of job ads per cell and year

	Minimum 5 ads	Minimum 10 ads	Minimum 20 ads	Minimum 50 ads
Job ads (log number)				
Entry-level ads	−0.074**	−0.093***	−0.094***	−0.094***
	(0.032)	(0.025)	(0.025)	(0.025)
Senior ads	−0.063**	−0.083***	−0.083***	−0.083***
	(0.028)	(0.020)	(0.020)	(0.020)
Entry-level ads				
Skill requirements	−0.046***	−0.047***	−0.047***	−0.048***
per ad (log)	(0.012)	(0.012)	(0.012)	(0.012)
Language	−1.10***	−1.10***	−1.11***	−1.12***
	(0.30)	(0.30)	(0.30)	(0.30)
Digital	−1.40***	−1.41***	−1.41***	−1.42***
	(0.33)	(0.33)	(0.33)	(0.33)
Analytical	−1.34***	−1.34***	−1.34***	−1.33***
	(0.37)	(0.37)	(0.37)	(0.38)
Social	−0.99***	−1.01***	−1.01***	−1.04***
	(0.29)	(0.28)	(0.28)	(0.28)
Interpersonal	−1.01**	−1.02**	−1.02**	−1.03**
	(0.43)	(0.44)	(0.44)	(0.44)
People management	−1.04***	−1.04***	−1.03***	−1.02***
	(0.22)	(0.22)	(0.22)	(0.22)
Writing	−0.01	−0.01	−0.01	−0.01
	(0.09)	(0.09)	(0.09)	(0.09)
Manual	−0.12	−0.13	−0.13	−0.13
	(0.30)	(0.30)	(0.30)	(0.30)
Entry-level ads minus senior ads in the same cell				
Skill requirements	0.008	0.007	0.007	0.007
per ad (log)	(0.007)	(0.007)	(0.007)	(0.006)
Language	0.21	0.21	0.20	0.20
	(0.21)	(0.21)	(0.21)	(0.21)
Digital	0.38	0.37	0.37	0.36
	(0.30)	(0.30)	(0.30)	(0.30)
Analytical	−0.31	−0.31	−0.31	−0.32
	(0.19)	(0.19)	(0.19)	(0.19)
Social	−0.25	−0.26	−0.26	−0.28
	(0.21)	(0.21)	(0.21)	(0.21)
Interpersonal	−0.03	−0.03	−0.03	−0.04
	(0.19)	(0.19)	(0.19)	(0.19)
People management	−0.27*	−0.27*	−0.26*	−0.26*
	(0.15)	(0.15)	(0.15)	(0.15)
Writing	0.04	0.04	0.04	0.04
	(0.08)	(0.08)	(0.08)	(0.08)
Manual	−0.25	−0.26	−0.26	−0.26
	(0.16)	(0.16)	(0.16)	(0.16)
Job ads				
Cells	8,016	7,203	6,245	4,866
Occupations	415	411	405	394
Ads covered (%)	99	98	98	98
Skill requirements				
Cells	2,743	2,516	2,240	1,787
Ads covered (%)	100	100	100	99

Notes: Columns: minimum number of job ads that state an experience requirement (job ads) or of entry-level ads and of senior ads (skill requirements) per cell and year; the paper uses ten. Estimates: change between the pre-period and the post-period per standard deviation of automation exposure to natural language processing, with country fixed effects, as in Tables 2 and 3, with the same outcomes as Table 4; standard errors clustered by occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent. Ads covered: share of the job ads that state an experience requirement (job ads) or of the entry-level ads and senior ads (skill requirements) that belong to the sample cells.

B Data processing and estimation

This appendix describes how the sample, the outcomes, the exposure measures and the estimates are constructed. All computations on individual job ads take place in the secure environment of the WIH; only counts aggregated to cells leave it.

B.1 Job ads

Each job ad has a unique identifier, a date on which it was first active, one or more locations and one or more sources. The year of a job ad is the year in which it was first active. A job ad with locations in several countries is assigned to the first of these countries in alphabetical order. Job ads are counted by country, four-digit occupation, industry section, year and source, separately for each experience band; each job ad is counted once, whatever the number of its locations and sources. Job ads without an occupation are excluded, and so are armed forces occupations.

A cell of the job-ad sample is a combination of country and four-digit occupation, summed over industries and sources. A cell enters the sample when it has at least ten job ads that state an experience requirement in every year from 2020 to 2025. The number of entry-level ads is the number of job ads in the bands "no experience" and "less than one year"; the number of senior ads is the number of job ads in the other bands; both enter in logs, with a floor of one job ad.

A continuous source is a source with at least 20 job ads in a country in every quarter between 2020 and 2025. For the robustness checks, a job ad comes from continuous sources when at least one of its sources is a continuous source in its country, and in the specification within source each cell is split by the job ad's first continuous source.

B.2 Skill requirements

The skill requirements come from the ESCO skills of each job ad. A skill requirement is a distinct ESCO skill in a job ad. Each ESCO skill belongs to at most one skill type by the rules of Table B.2; skills of no skill type count toward the number of skill requirements but toward no skill type. Job ads are counted by country, three-digit occupation, industry section, half-year, source and experience group: entry-level ads and senior ads, the latter requiring one year of experience or more. For each group, I count the job ads, the skill requirements, and the job ads that require at least one skill of each skill type. Half-years are summed to years.

A cell of the skill sample is a combination of country and three-digit occupation with at least ten entry-level ads and ten senior ads in every year from 2020 to 2025. The share of job ads that require skill type g is $100\,E^g_{ct}/E_{ct}$, where E_{ct} is the number of job ads of the experience group and E^g_{ct} the number that require at least one skill of type g. The log number of skill requirements per job ad is $\ln\big(\max(K_{ct},1)/E_{ct}\big)$, where K_{ct} is the number of skill requirements summed over the job ads of the group.

For the comparison with education requirements, the shares of entry-level ads and of senior ads that require tertiary education and low education are computed for each skill cell and year among the job ads that state an education requirement, summed over the four-digit occupations of the three-digit occupation. Their changes between the pre-period and the post-period enter

the pooled regression as additional regressors: the two shares of entry-level ads for entry-level ads, the two shares of senior ads for senior ads, and all four for the difference. Cells in which entry-level ads or senior ads state no education requirement in some year are dropped, which leaves 2,476 of the 2,516 cells; the estimates without the additional regressors refer to the same cells.

For the comparison within education levels, job ads are counted in addition by the education level that they require (low, mid-level or tertiary; job ads that state no education requirement are excluded), and each skill cell is split by education level. A cell of country, three-digit occupation and education level enters the sample when it has at least ten entry-level ads and ten senior ads in every year from 2020 to 2025: 4,123 cells in 122 occupations. The outcomes are those of the skill sample and the log number of job ads; the regressions are estimated separately for each education level with country fixed effects, and pooled over education levels with country $\times$ education fixed effects. Before job ads without an education requirement are excluded, these counts summed over education levels equal the counts of the skill sample exactly.

B.3 Exposure

Automation exposure is assigned to cells by four-digit occupation. For the skill sample, the automation exposure of a three-digit occupation is the average of the automation exposure of its four-digit occupations, weighted by their job ads that state an experience requirement in 2020–2022, summed over all countries. Exposure is standardized across the cells of each sample, without weights. The fifths of automation exposure are formed by sorting cells by exposure and cutting the cumulative pre-period job ads that state an experience requirement into five equal parts.

Augmentation exposure is constructed from the same patents and the same matching procedure, applied to texts that describe what an occupation produces rather than what its workers do: the occupational job titles of Tijdens (2023), each combined with each industry section and preceded by "a new product, device, or technology produced or used by a worker in:", with the matching threshold computed separately for these texts. The numbers of matches are counted for each four-digit occupation and industry section, and excess augmentation exposure follows Equation (3). The augmentation exposure of a three-digit occupation is the average over its four-digit occupations and industry sections, weighted by their job ads that state an experience requirement in 2020–2022.

B.4 Estimation

All estimates are weighted least-squares regressions. The pooled estimate regresses the difference between the cell's post-period average and pre-period average on the exposure indicators or on standardized exposure, after subtracting weighted means within each country (within each country $\times$ industry pair or country $\times$ source pair in the corresponding robustness checks). The event-study estimate for year t uses the difference between the cell's outcome in year t and in 2022. The weight of a cell is its average number of job ads that state an experience requirement in the pre-period (entry-level ads plus senior ads in the skill sample). Standard errors are

cluster-robust by occupation,

$$\hat{V} = (X'WX)^{-1}\Big(\sum_g s_g s_g'\Big)(X'WX)^{-1} \cdot \frac{G}{G-1} \cdot \frac{N_c - 1}{N_c - K},$$

where X contains the demeaned regressors, W the weights, s_g the sum of the weighted scores over the cells of occupation g, G the number of occupations, N_c the number of cells and K the number of regressors.

For the comparison with augmentation exposure, both exposures are standardized and enter one regression. The estimate with automation exposure alone equals $\hat{\beta}_{aut} + \hat{\delta}\,\hat{\beta}_{aug}$, where $\hat{\beta}_{aut}$ and $\hat{\beta}_{aug}$ are the coefficients of the joint regression and $\hat{\delta}$ is the coefficient of a regression of standardized augmentation exposure on standardized automation exposure with the same fixed effects and weights; this omitted-variable formula holds exactly in the sample.

Table B.1: Entry-level ads, senior ads and the entry-level share of all job ads, 2020–2026

	Entry-level ads	Senior ads	Entry-level share (%)		UK share of ads
	(million)	(million)	Actual	Country weights of 2021–22	stating experience (%)
2020	10.4	9.7	51.7	48.4	40.8
2021	11.9	10.0	54.3	54.2	32.2
2022	13.0	11.2	53.8	53.9	30.1
2023	12.5	12.5	50.0	53.3	23.5
2024	10.7	11.5	48.2	53.3	11.7
2025	8.8	10.0	47.1	53.8	9.9
2026 H1	3.9	5.0	43.9	53.1	5.0

Notes: All job ads in the 27 member states of the European Union and the United Kingdom, before the selection of cells. Entry-level share: entry-level ads as a share of job ads that state an experience requirement. Country weights of 2021–22: average of the countries' entry-level shares, weighted by their job ads that state an experience requirement in 2021–2022. 2026 H1: January to June 2026.

Table B.2: Skill types: rules applied to the ESCO classification

Skill type	ESCO groups and flags (applied in this order)	Concepts
Language	language flag or language pillar; S1.15 using more than one language; T1.1 mastering languages	401
Writing	S1.13 writing and composing; labels containing write, draft, compose, edit, proofread or copywrite (not music)	183
Social	S1.1 negotiating; S1.2 liaising and networking; S1.4 presenting information; S1.7 obtaining information verbally; S1.8 working with others; T4 and T4.1 communicating; T4.3 collaborating in teams and networks	641
Interpersonal	S1.3 teaching and training; S1.5 advising and consulting; S1.6 promoting, selling and purchasing; S3.1 counseling; S3.4 providing information and support to the public and clients; S3.6 providing general personal care; T4.2 supporting others	1,127
People management	S4.5 leading and motivating; S4.6 building and developing teams; S4.7 recruiting and hiring; S4.8 supervising people; T4.4 leading others	238
Digital	S5 working with computers; ICT flag; T1.3 working with digital devices and applications	1,092
Analytical	S1.9 solving problems; S2.1 conducting studies, investigations and examinations; S2.6 calculating and estimating; S2.7 analyzing and evaluating information and data; T2 thinking skills and competences	857
Manual	S6 handling and moving; S7 constructing; S8 working with machinery and specialized equipment; T5 physical and manual skills and competences	3,138
No skill type	everything else (mostly knowledge of particular fields)	6,312
Total		13,989

Notes: Codes and labels of the ESCO skills hierarchy (S: skills, T: transversal skills and competences). A concept that meets several rules belongs to the first skill type in the order of the table. Concepts: number of ESCO skills, knowledge items and competences of the skill type.

Table B.3: Patents by AI type and vintage

	Breakthrough patents 1976–2015		All patents 2010–2020	
	Patents	Share, %	Patents	Share, %
All patents	526,233	100.0	3,210,246	100.0
All AI	81,914	15.6	538,932	16.8
Machine learning	10,940	2.1	64,637	2.0
Evolutionary computation	2,724	0.5	25,480	0.8
Natural language processing	10,421	2.0	86,811	2.7
Speech	6,422	1.2	35,831	1.1
Computer vision	21,133	4.0	155,971	4.9
Planning and control	34,454	6.5	212,844	6.6
Knowledge processing	16,759	3.2	128,155	4.0
AI hardware	28,761	5.5	196,646	6.1
Industrial robots (placebo)	866	0.2	7,522	0.2

Notes: Utility patents granted by the USPTO. AI types: USPTO AI Patent Dataset (Pairolero et al., 2024), predicted probability of at least 0.86; a patent with several labels counts toward each AI type. Breakthrough patents: indicator of Kelly et al. (2021), extended to patents granted until 2015.

Table B.4: Job ads and skill requirements after ChatGPT, by AI type

	(1) All AI	(2) Machine learning	(3) Evolutionary computation	(4) Natural language processing	(5) Speech	(6) Computer vision	(7) Planning and control	(8) Knowledge processing	(9) AI hardware	(10) Industrial robots (placebo)
Job ads (log number)										
Entry-level ads	−0.099***	−0.095***	−0.092***	−0.093***	−0.062***	−0.008	−0.095***	−0.104***	−0.103***	−0.005
	(0.021)	(0.021)	(0.020)	(0.025)	(0.020)	(0.023)	(0.021)	(0.023)	(0.019)	(0.022)
Senior ads	−0.090***	−0.077***	−0.077***	−0.083***	−0.052***	−0.004	−0.088***	−0.094***	−0.090***	0.001
	(0.017)	(0.018)	(0.016)	(0.020)	(0.017)	(0.021)	(0.016)	(0.017)	(0.016)	(0.017)
Entry-level ads										
Skill requirements	−0.052***	−0.051***	−0.049***	−0.047***	−0.033**	−0.018	−0.050***	−0.050***	−0.046***	0.016
per ad (log)	(0.009)	(0.008)	(0.007)	(0.012)	(0.014)	(0.015)	(0.009)	(0.009)	(0.007)	(0.011)
Language	−1.14***	−1.08***	−1.10***	−1.10***	−0.91***	−0.43	−1.16***	−1.13***	−0.86***	0.95***
	(0.30)	(0.28)	(0.27)	(0.30)	(0.31)	(0.34)	(0.27)	(0.29)	(0.33)	(0.29)
Digital	−1.54***	−1.41***	−1.31***	−1.41***	−0.93**	−0.27	−1.50***	−1.52***	−1.43***	0.36
	(0.26)	(0.25)	(0.23)	(0.33)	(0.41)	(0.48)	(0.27)	(0.27)	(0.25)	(0.27)
Analytical	−1.56***	−1.36***	−1.48***	−1.34***	−0.74*	0.16	−1.61***	−1.60***	−1.46***	0.09
	(0.29)	(0.33)	(0.26)	(0.37)	(0.40)	(0.45)	(0.30)	(0.31)	(0.24)	(0.40)
Social	−0.95***	−0.83***	−0.84***	−1.01***	−0.83***	0.08	−1.01***	−1.01***	−0.80***	0.15
	(0.25)	(0.24)	(0.20)	(0.28)	(0.31)	(0.28)	(0.25)	(0.26)	(0.21)	(0.28)
Interpersonal	−1.27***	−1.28***	−1.27***	−1.02**	−0.46	−0.20	−1.20***	−1.25***	−1.32***	0.09
	(0.32)	(0.29)	(0.24)	(0.44)	(0.49)	(0.38)	(0.34)	(0.35)	(0.23)	(0.42)
People management	−1.15***	−1.11***	−1.07***	−1.04***	−0.78***	0.09	−1.16***	−1.20***	−1.07***	0.23
	(0.18)	(0.19)	(0.16)	(0.22)	(0.27)	(0.34)	(0.19)	(0.19)	(0.20)	(0.27)
Writing	−0.05	−0.04	−0.05	−0.01	0.04	−0.27*	−0.01	−0.01	−0.07	−0.07
	(0.09)	(0.08)	(0.09)	(0.09)	(0.09)	(0.16)	(0.08)	(0.09)	(0.09)	(0.07)
Manual	−0.49*	−0.50*	−0.57**	−0.13	0.28	−0.43	−0.44	−0.42	−0.60**	−0.10
	(0.29)	(0.29)	(0.24)	(0.30)	(0.26)	(0.31)	(0.29)	(0.30)	(0.25)	(0.20)
Entry-level ads minus senior ads in the same cell										
Skill requirements	0.007	0.005	0.006	0.007	0.004	−0.008	0.009	0.009	0.009	0.004
per ad (log)	(0.007)	(0.008)	(0.006)	(0.007)	(0.005)	(0.006)	(0.006)	(0.007)	(0.007)	(0.005)
Language	0.18	0.12	0.06	0.21	0.20	−0.23	0.23	0.24	0.21	0.26
	(0.18)	(0.19)	(0.19)	(0.21)	(0.20)	(0.17)	(0.17)	(0.18)	(0.16)	(0.25)
Digital	0.20	0.20	0.16	0.37	0.47*	−0.13	0.24	0.26	0.13	−0.35
	(0.31)	(0.38)	(0.31)	(0.30)	(0.25)	(0.30)	(0.28)	(0.30)	(0.27)	(0.26)
Analytical	−0.32*	−0.23	−0.23	−0.31	−0.12	−0.24	−0.26	−0.32	−0.25	0.16
	(0.19)	(0.18)	(0.17)	(0.19)	(0.21)	(0.26)	(0.20)	(0.20)	(0.18)	(0.17)
Social	−0.19	−0.15	−0.09	−0.26	−0.20	−0.21	−0.11	−0.16	−0.18	0.15
	(0.18)	(0.20)	(0.17)	(0.21)	(0.23)	(0.19)	(0.18)	(0.18)	(0.18)	(0.16)
Interpersonal	0.12	0.10	0.14	−0.03	−0.18	0.01	0.16	0.14	0.23	0.20
	(0.19)	(0.18)	(0.17)	(0.19)	(0.20)	(0.21)	(0.20)	(0.19)	(0.19)	(0.16)
People management	−0.40***	−0.34***	−0.35***	−0.27*	−0.09	0.02	−0.37***	−0.39***	−0.41***	0.03
	(0.12)	(0.12)	(0.11)	(0.15)	(0.18)	(0.19)	(0.13)	(0.13)	(0.12)	(0.13)
Writing	0.09	0.07	0.09	0.04	−0.04	−0.06	0.09	0.09	0.13	0.07
	(0.08)	(0.07)	(0.07)	(0.08)	(0.06)	(0.07)	(0.08)	(0.08)	(0.08)	(0.06)
Manual	−0.29*	−0.37**	−0.31**	−0.26	−0.21	−0.39*	−0.24	−0.26	−0.21	0.21**
	(0.18)	(0.18)	(0.16)	(0.16)	(0.15)	(0.21)	(0.17)	(0.17)	(0.17)	(0.09)

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of automation exposure to each AI type, estimated separately for each AI type, with country fixed effects; every column reports the same outcomes as Table 4. Job ads: log number of entry-level ads and of senior ads; cells of country × four-digit occupation, standard errors clustered by four-digit occupation. Skill requirements: log number of skill requirements per job ad and share of job ads that require at least one skill of the skill type (percentage points), for entry-level ads and for entry-level ads minus senior ads in the same cell; cells of country × three-digit occupation, standard errors clustered by three-digit occupation. AI types as labeled in the USPTO AI Patent Dataset; breakthrough patents granted 1976–2015. Standard errors in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

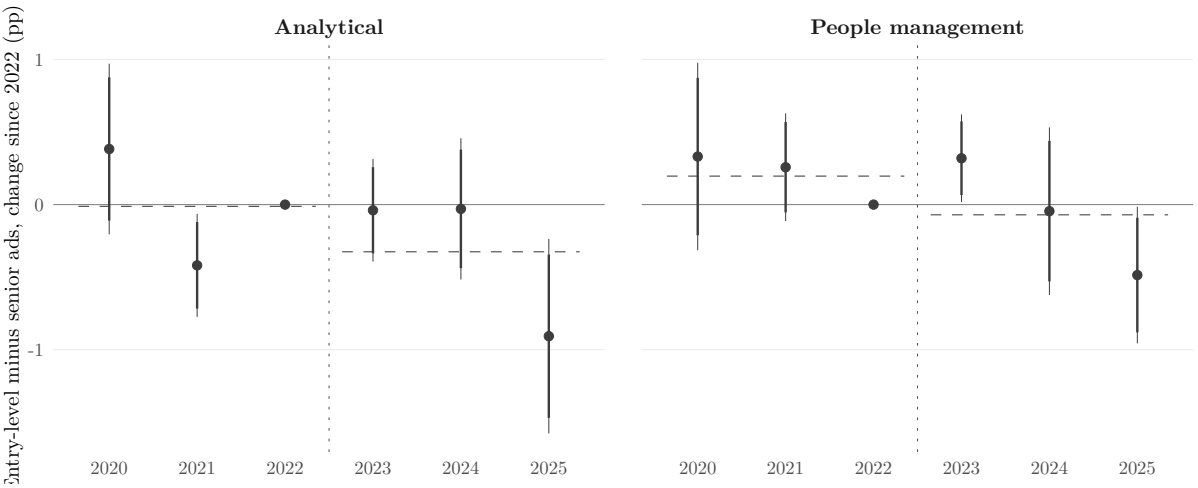


Figure B.1: Entry-level ads minus senior ads: analytical and people-management skill requirements over time.

Notes: Change between 2022 and each year in the share of entry-level ads that require the skill type minus the same share in senior ads of the same cell, per standard deviation of automation exposure to natural language processing, with country fixed effects. Annual data; dashed lines are the averages of the yearly coefficients in the pre-period (including the base year 2022) and in the post-period, whose distance equals the pooled estimate in Table 3, column 3. Thick bars are 90 percent and thin bars 95 percent confidence intervals, clustered by three-digit occupation.

C Additional results

This appendix reports the education requirements of job ads, the estimates behind Figures 4 and 5, and two checks of Section 5: senior ads defined as job ads that require four years of experience or more, and automation exposure together with augmentation exposure.

C.1 Education requirements

Automation exposure also changed the education that job ads require (Table C.1). Per standard deviation of automation exposure, the share of senior ads that require tertiary education fell by 0.59 percentage points and the share that requires mid-level education rose by 0.68 percentage points. In entry-level ads, the share that requires mid-level education rose by 0.55 percentage points and the share that requires low education fell by 0.42 percentage points; the share that requires tertiary education did not change. Tertiary requirements therefore fell more in senior ads than in entry-level ads, by 0.46 percentage points. For senior ads, automation exposure to industrial robots predicts no change. For entry-level ads, it predicts a smaller shift in the same direction, significant only at the 10 percent level, so the shift in entry-level ads is less clearly specific to generative AI than the shift in senior ads.

The number of entry-level ads fell at every education level, but by different amounts. Per standard deviation of automation exposure, entry-level ads that require tertiary education fell by 9.8 log points, entry-level ads that require low education by 13.0 log points, and entry-level ads that require mid-level education by 7.7 log points. Entry-level jobs open to workers with mid-level education, the level of vocational education and apprenticeships, therefore declined the least. By Proposition 2, applied to education instead of skills, generative AI automated the

tasks for which firms asked for tertiary education more than proportionally, consistent with the decline of analytical skill requirements.

Table C.1: Education requirements after ChatGPT, per standard deviation of automation exposure

	Natural language processing			Industrial robots (placebo)	
	(1) Entry-level ads	(2) Senior ads	(3) Difference	(4) Entry-level ads	(5) Senior ads
Share of ads that require the education level (percentage points)					
Tertiary (ISCED 5–8)	−0.13	−0.59***	0.46**	−0.46*	−0.09
	(0.27)	(0.19)	(0.19)	(0.27)	(0.15)
Mid-level (ISCED 3–4)	0.55**	0.68***	−0.13	0.34*	0.20
	(0.22)	(0.23)	(0.15)	(0.19)	(0.19)
Low (ISCED 0–2)	−0.42**	−0.09	−0.33**	0.12	−0.11
	(0.21)	(0.14)	(0.16)	(0.26)	(0.14)
Number of ads that require the education level (log)					
Tertiary (ISCED 5–8)	−0.098***	−0.114***	0.015	0.005	0.013
	(0.024)	(0.019)	(0.012)	(0.024)	(0.017)
Mid-level (ISCED 3–4)	−0.077***	−0.046**	−0.031**	0.007	−0.000
	(0.025)	(0.022)	(0.013)	(0.022)	(0.021)
Low (ISCED 0–2)	−0.130***	−0.083***	−0.047**	0.022	−0.016
	(0.031)	(0.030)	(0.019)	(0.036)	(0.029)
Cells			4,853		
Occupations (clusters)			396		

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of automation exposure, with country fixed effects. Upper panel: share of job ads that state an education requirement and require the education level, in percentage points. Lower panel: log number of job ads that require the education level. Columns (1) and (4): entry-level ads; columns (2) and (5): senior ads; column (3): entry-level ads minus senior ads in the same cell. Mid-level education (ISCED 3–4) includes vocational education and apprenticeships. Cells: country × four-digit occupation with at least ten entry-level ads and ten senior ads that state an education requirement in every year, weighted by their pre-period job ads that state an education requirement. Standard errors clustered by four-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

C.2 Job ads and skill requirements by education level

Table C.2 reports the estimates shown in Figures 4 and 5, separately for job ads that require tertiary, mid-level and low education.

Table C.2: Job ads and skill requirements after ChatGPT by education level, per standard deviation of automation exposure

	Tertiary education			Mid-level education			Low education		
	(1) Entry-level ads	(2) Senior ads	(3) Difference	(4) Entry-level ads	(5) Senior ads	(6) Difference	(7) Entry-level ads	(8) Senior ads	(9) Difference
Job ads	−0.091**	−0.116***	0.025	−0.071***	−0.041*	−0.030***	−0.079*	−0.039	−0.040
(log number)	(0.044)	(0.030)	(0.018)	(0.022)	(0.021)	(0.009)	(0.044)	(0.050)	(0.029)
Skill requirements	−0.046***	−0.040***	−0.006	−0.042***	−0.059***	0.017**	−0.070***	−0.057**	−0.013
per ad (log)	(0.010)	(0.008)	(0.005)	(0.014)	(0.014)	(0.008)	(0.020)	(0.024)	(0.028)
Share of ads that require the skill type (percentage points)									
Language	−1.19**	−1.33***	0.14	−1.07***	−1.33***	0.26*	−1.85**	−0.41	−1.44**
	(0.53)	(0.40)	(0.39)	(0.24)	(0.23)	(0.15)	(0.75)	(0.45)	(0.65)
Digital	−0.84**	−1.52***	0.68	−1.99***	−2.09***	0.10	−1.07***	−0.87*	−0.21
	(0.40)	(0.34)	(0.50)	(0.38)	(0.34)	(0.26)	(0.36)	(0.52)	(0.44)
Analytical	−0.90***	−0.55*	−0.36	−1.67***	−1.31***	−0.36	−0.20	−0.69	0.50
	(0.31)	(0.31)	(0.30)	(0.49)	(0.47)	(0.27)	(0.34)	(0.46)	(0.31)
Social	−0.89**	−0.21	−0.67	−1.18***	−1.13***	−0.05	−1.24***	−0.45	−0.80
	(0.44)	(0.40)	(0.43)	(0.36)	(0.31)	(0.27)	(0.38)	(0.51)	(0.59)
Interpersonal	−0.86	−0.80**	−0.06	−1.13**	−1.11**	−0.03	−0.04	−0.13	0.09
	(0.56)	(0.39)	(0.32)	(0.57)	(0.52)	(0.26)	(0.53)	(0.36)	(0.59)
People management	−0.62***	−0.20	−0.42	−1.31***	−1.16***	−0.15	0.42	−0.71**	1.13**
	(0.24)	(0.16)	(0.30)	(0.27)	(0.28)	(0.18)	(0.27)	(0.35)	(0.49)
Writing	−0.07	−0.10	0.03	0.13	0.10	0.03	−0.21	−0.20	−0.01
	(0.06)	(0.08)	(0.07)	(0.09)	(0.12)	(0.08)	(0.20)	(0.24)	(0.07)
Manual	−0.39	0.13	−0.52	0.38	0.36	0.01	0.04	−0.32	0.36
	(0.36)	(0.20)	(0.32)	(0.28)	(0.31)	(0.20)	(0.39)	(0.28)	(0.33)
Cells		1,397			1,973			753	
Occupations		119			119			103	

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of automation exposure to natural language processing, with country fixed effects, estimated separately for each education level that the job ads require. Cells: country × three-digit occupation × education level, with at least ten entry-level ads and ten senior ads in every year, weighted by their pre-period entry-level ads and senior ads. Columns (3), (6) and (9): entry-level ads minus senior ads in the same cell. First row: log number of job ads. Second row: log number of skill requirements per job ad. Other rows: share of job ads that require at least one skill of the skill type, in percentage points. Tertiary education: ISCED 5–8; mid-level education: ISCED 3–4, including vocational education and apprenticeships; low education: ISCED 0–2. Standard errors clustered by three-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

C.3 Senior ads that require four years of experience or more

Senior ads, as defined in the paper, include job ads that require one or two years of experience, which may differ little from entry-level ads. Tables C.3 and C.4 therefore compare entry-level ads with job ads that require four years of experience or more, the WIH bands of four to six years and of more than six years, which make up 19 percent of the pre-period job ads that state an experience requirement. Job ads that require one to four years form a middle group. Cells are combinations of country and three-digit occupation, because the four-digit job-ad panel does not separate the bands above two years.

The number of entry-level ads fell relative to the number of job ads that require four years or more, by 2.9 log points per standard deviation of automation exposure (Table C.3, column 4), also without the countries whose coding of the experience requirement changed and in cells with at least ten entry-level ads and ten job ads that require four years or more. The yearly estimates,

however, show that the relative decline happened before 2022: in the top fifth of automation exposure relative to the bottom fifth, the difference between the log numbers of entry-level ads and of job ads that require four years or more was 22 and 14 log points higher in 2020 and 2021 than in 2022, and 3, −2 and 13 log points higher in 2023, 2024 and 2025. The relative decline of entry-level ads therefore cannot be attributed to the release of ChatGPT. Automation exposure to industrial robots predicts no change.

For skill requirements, Table C.4 uses the 2,030 cells with at least ten entry-level ads, ten job ads that require one to four years and ten job ads that require four years or more in every year, so that all comparisons use the same cells and the same weights. Entry-level ads and job ads that require four years or more lost the same skill types (columns 1 and 2). The differences between them (column 3) are no larger than with the definition of the paper (column 4): −0.28 against −0.31 percentage points for analytical skill requirements and −0.27 in both for people-management skill requirements, with larger standard errors. Job ads that require one to four years differ from both groups (columns 5 and 6): their language skill requirements fell more and their social skill requirements less than those of entry-level ads and of job ads that require four years or more. The differences between entry-level ads and job ads that require four years or more have the same signs for every skill type without the countries whose coding of the experience requirement changed, on all 2,083 cells with entry-level ads and job ads that require four years or more, and within education levels (columns 7 to 9). A sharper definition of senior ads therefore does not reveal a larger difference between entry-level ads and senior ads.

Table C.3: Senior ads that require four years of experience or more: job ads

	(1) Entry-level ads (log)	(2) Ads requiring 1–4 years (log)	(3) Ads requiring 4+ years (log)	(4) Entry-level minus 4+ (log)	(5) Entry-level share (pp)
Per standard deviation of automation exposure					
Natural language processing	−0.089***	−0.084***	−0.060**	−0.029***	−0.39*
	(0.029)	(0.029)	(0.026)	(0.009)	(0.21)
without coding breaks	−0.079**	−0.073**	−0.045	−0.035***	−0.51*
	(0.033)	(0.034)	(0.029)	(0.013)	(0.26)
at least ten entry-level and 4+ ads	−0.091***	−0.085***	−0.063**	−0.028***	−0.38*
	(0.029)	(0.029)	(0.026)	(0.009)	(0.21)
Industrial robots (placebo)	−0.010	0.001	−0.009	−0.001	−0.24
	(0.030)	(0.030)	(0.027)	(0.008)	(0.17)
Top fifth relative to bottom fifth, natural language processing					
Pooled	−0.200*	−0.191	−0.126	−0.073**	−1.09
	(0.105)	(0.119)	(0.097)	(0.030)	(0.69)
2020 relative to 2022	0.096	−0.094	−0.123*	0.219**	3.59**
	(0.076)	(0.078)	(0.072)	(0.107)	(1.79)
2021 relative to 2022	−0.075	−0.143*	−0.219***	0.143***	2.58***
	(0.064)	(0.086)	(0.069)	(0.043)	(0.78)
2023 relative to 2022	−0.118***	−0.249***	−0.145***	0.028	1.23*
	(0.040)	(0.092)	(0.056)	(0.043)	(0.69)
2024 relative to 2022	−0.233**	−0.269**	−0.216***	−0.017	−0.39
	(0.117)	(0.125)	(0.079)	(0.063)	(1.33)
2025 relative to 2022	−0.227	−0.291**	−0.360***	0.132	2.06
	(0.150)	(0.138)	(0.104)	(0.090)	(1.53)
Cells			2,955		

Notes: Change between the pre-period (2020–22) and the post-period (2023–25), or between 2022 and the year, in the log number of entry-level ads, of job ads that require one to four years of experience and of job ads that require four years or more (WIH bands of four to six years and more than six years); column (4): entry-level ads minus job ads that require four years or more (log); column (5): entry-level ads as a share of entry-level ads and job ads that require four years or more (percentage points). Cells: country × three-digit occupation with at least ten job ads that state an experience requirement in every year, with country fixed effects, weighted by pre-period job ads that state an experience requirement. Countries with coding breaks: countries whose share of job ads that require four years or more among job ads that state an experience requirement changes by more than ten percentage points from one year to the next (Czechia, Germany, the Netherlands, Poland). Standard errors clustered by three-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

Table C.4: Senior ads that require four years of experience or more: skill requirements

	Entry-level and 4+ ads			Other differences, same cells			Entry-level minus 4+, checks		
	(1) Entry-level ads	(2) Ads requiring 4+ years	(3) Difference	(4) Entry-level minus 1+	(5) Entry-level minus 1–4	(6) 1–4 minus 4+ years	(7) Without coding breaks	(8) All cells	(9) Within education
Skill requirements per ad (log)	−0.048***	−0.056***	0.009	0.007	0.000	0.008	0.000	0.010	0.007
	(0.012)	(0.014)	(0.008)	(0.007)	(0.006)	(0.005)	(0.006)	(0.009)	(0.008)
Share of ads that require the skill type (percentage points)									
Language	−1.12***	−1.09***	−0.02	0.20	0.49**	−0.51***	−0.35	−0.06	−0.07
	(0.30)	(0.27)	(0.22)	(0.21)	(0.23)	(0.17)	(0.24)	(0.22)	(0.22)
Digital	−1.41***	−1.64***	0.24	0.37	0.25	−0.02	0.15	0.15	0.08
	(0.33)	(0.30)	(0.29)	(0.30)	(0.34)	(0.34)	(0.38)	(0.30)	(0.28)
Analytical	−1.33***	−1.06**	−0.28	−0.31	−0.43*	0.16	−0.07	−0.30	−0.41
	(0.37)	(0.41)	(0.24)	(0.19)	(0.23)	(0.24)	(0.29)	(0.26)	(0.28)
Social	−1.02***	−0.86***	−0.16	−0.27	−0.61***	0.45*	−0.06	−0.09	−0.28
	(0.28)	(0.28)	(0.25)	(0.21)	(0.23)	(0.24)	(0.23)	(0.26)	(0.26)
Interpersonal	−1.03**	−0.81*	−0.22	−0.04	0.18	−0.40*	−0.19	−0.25	−0.39**
	(0.44)	(0.48)	(0.21)	(0.19)	(0.23)	(0.23)	(0.24)	(0.22)	(0.20)
People management	−1.04***	−0.77***	−0.27	−0.27*	−0.25	−0.02	−0.38*	−0.21	−0.23
	(0.22)	(0.21)	(0.19)	(0.15)	(0.18)	(0.20)	(0.23)	(0.21)	(0.22)
Writing	−0.01	−0.02	0.01	0.04	0.04	−0.03	0.05	0.03	0.02
	(0.09)	(0.14)	(0.09)	(0.08)	(0.05)	(0.06)	(0.12)	(0.11)	(0.10)
Manual	−0.13	0.05	−0.18	−0.25	−0.24	0.07	−0.47**	−0.29	−0.23
	(0.30)	(0.30)	(0.18)	(0.16)	(0.20)	(0.18)	(0.21)	(0.20)	(0.20)
Cells	2,030						1,640	2,083	3,069

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of automation exposure to natural language processing, with country fixed effects. Columns (1) to (6): the 2,030 cells (country × three-digit occupation) with at least ten entry-level ads, ten job ads that require one to four years and ten job ads that require four years or more in every year, all weighted by their pre-period job ads that state an experience requirement. Column (4): entry-level ads minus senior ads as defined in the paper (one year or more); column (5): entry-level ads minus job ads that require one to four years; column (6): job ads that require one to four years minus job ads that require four years or more. Columns (7) to (9): entry-level ads minus job ads that require four years or more, without countries with coding breaks (Table C.3), on all cells with at least ten entry-level ads and ten job ads that require four years or more, and within education levels (cells split by the education level that the job ads require, with country × education fixed effects). First row: log number of skill requirements per job ad. Standard errors clustered by three-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.

C.4 Automation exposure and augmentation exposure

Automation exposure measures whether patents can perform the tasks of an occupation. The same patents can also expand what an occupation produces and create new tasks in it (Autor et al., 2024). I measure this augmentation exposure by matching the patents of each AI type to texts that describe what an occupation produces in an industry (Appendix B). Across three-digit occupations, augmentation exposure to natural language processing is correlated with automation exposure to natural language processing at 0.82, with weights of pre-period job ads. An estimate with automation exposure alone therefore also captures part of the effect of augmentation exposure.

Table C.5 shows how large this part is. Column (1) repeats the estimates with automation exposure alone, on cells of country and three-digit occupation, and column (2) uses augmentation exposure alone. Columns (3) and (4) include both exposures in one regression. Column (5) is the difference between columns (1) and (3); it equals the coefficient of augmentation exposure in column (4) times 0.82, the slope of augmentation exposure on automation exposure.

The automation estimates of the main text are lower bounds for skill requirements and for education requirements. Holding augmentation exposure fixed, automation exposure lowers the shares of entry-level ads that require language, digital, analytical, interpersonal and people-management skills by 1.4 to 2.7 percentage points per standard deviation, instead of 1.0 to 1.4, and the number of skill requirements per entry-level ad by 6.9 instead of 4.7 percent. Augmentation exposure, holding automation exposure fixed, works in the opposite direction: it raises the shares of entry-level ads that require digital and language skills, by 1.54 and 0.99 percentage points, and the share that requires manual skills. Holding augmentation exposure fixed, automation exposure lowers the shares of entry-level ads and of senior ads that require tertiary education, by 0.73 and 0.86 percentage points, whereas augmentation exposure raises them. The difference between entry-level ads and senior ads in people-management skill requirements grows from -0.27 to -0.45 percentage points and becomes significant at the 5 percent level. The number of job ads does not change: holding automation exposure fixed, augmentation exposure is unrelated to the number of entry-level ads and of senior ads.

Two caveats apply. The separate effects of the two exposures are identified from occupations in which they differ: among the largest occupations, health care assistants, ICT user support technicians, cleaners and helpers, teachers' aides and waiters have more augmentation exposure than their automation exposure predicts, and draughtspersons, personnel and careers professionals, accountants, electrical mechanics and fitters, and electronics engineers have less. These groups differ in other ways as well. Moreover, augmentation exposure to industrial robots, the placebo technology, also predicts higher skill requirements in entry-level ads, by 0.7 to 1.1 percentage points per standard deviation for language, digital, analytical, interpersonal and people-management skills, so part of what augmentation exposure measures may be a contrast between technical and other occupations rather than new tasks created by AI.

Table C.5: Automation exposure and augmentation exposure of three-digit occupations

			Both exposures		
	(1) Automation exposure alone	(2) Augmentation exposure alone	(3) Automation exposure	(4) Augmentation exposure	(5) Difference (1) − (3)
Job ads (log number)					
Entry-level ads	−0.089***	−0.071**	−0.092***	0.003	0.002
	(0.029)	(0.031)	(0.032)	(0.037)	
Senior ads	−0.078***	−0.066**	−0.068**	−0.012	−0.010
	(0.025)	(0.028)	(0.027)	(0.033)	
Entry-level ads					
Skill requirements	−0.047***	−0.028**	−0.069***	0.027*	0.022
per ad (log)	(0.012)	(0.012)	(0.014)	(0.014)	
Language	−1.10***	−0.55*	−1.92***	0.99***	0.82
	(0.30)	(0.30)	(0.32)	(0.36)	
Digital	−1.41***	−0.61*	−2.67***	1.54***	1.27
	(0.33)	(0.34)	(0.37)	(0.44)	
Analytical	−1.34***	−0.91**	−1.74***	0.49	0.40
	(0.37)	(0.41)	(0.45)	(0.50)	
Social	−1.01***	−0.81***	−0.99**	−0.01	−0.01
	(0.28)	(0.27)	(0.47)	(0.43)	
Interpersonal	−1.02**	−0.67	−1.38***	0.44	0.36
	(0.44)	(0.46)	(0.48)	(0.53)	
People management	−1.04***	−0.67***	−1.43***	0.47	0.39
	(0.22)	(0.21)	(0.36)	(0.34)	
Writing	−0.01	−0.08	0.14	−0.19	−0.16
	(0.09)	(0.10)	(0.09)	(0.13)	
Manual	−0.13	0.26	−1.01*	1.08**	0.89
	(0.30)	(0.28)	(0.56)	(0.54)	
Entry-level ads minus senior ads in the same cell					
Skill requirements	0.007	0.009	−0.000	0.009	0.007
per ad (log)	(0.007)	(0.006)	(0.011)	(0.010)	
Language	0.21	0.28	−0.06	0.33	0.27
	(0.21)	(0.23)	(0.23)	(0.30)	
Digital	0.37	0.56**	−0.25	0.76	0.62
	(0.30)	(0.22)	(0.59)	(0.53)	
Analytical	−0.31	−0.19	−0.47	0.20	0.16
	(0.19)	(0.19)	(0.37)	(0.36)	
Social	−0.26	−0.26	−0.11	−0.17	−0.14
	(0.21)	(0.22)	(0.39)	(0.41)	
Interpersonal	−0.03	−0.16	0.29	−0.40	−0.33
	(0.19)	(0.20)	(0.35)	(0.37)	
People management	−0.27*	−0.14	−0.45**	0.22	0.18
	(0.15)	(0.15)	(0.23)	(0.22)	
Writing	0.04	0.01	0.08	−0.05	−0.04
	(0.08)	(0.09)	(0.06)	(0.10)	
Manual	−0.26	−0.04	−0.67**	0.50*	0.41
	(0.16)	(0.11)	(0.33)	(0.30)	
Share of job ads that require the education level (percentage points)					
Tertiary, entry-level ads	−0.08	0.21	−0.73**	0.79**	0.65
	(0.28)	(0.28)	(0.35)	(0.37)	
Tertiary, senior ads	−0.50***	−0.25	−0.86***	0.43*	0.36
	(0.16)	(0.18)	(0.22)	(0.25)	
Mid-level, entry-level ads	0.51*	0.30	0.76***	−0.31	−0.26
	(0.30)	(0.34)	(0.29)	(0.38)	
Mid-level, senior ads	0.65**	0.42	0.92***	−0.32	−0.26
	(0.33)	(0.37)	(0.26)	(0.36)	
Low, entry-level ads	−0.43*	−0.51*	−0.04	−0.48*	−0.39
	(0.26)	(0.26)	(0.21)	(0.28)	
Low, senior ads	−0.15	−0.16	−0.06	−0.12	−0.10
	(0.22)	(0.24)	(0.15)	(0.24)	
Cells					
Job ads			2,955		
Skill requirements			2,516		
Education requirements			2,388		

Notes: Change between the pre-period (2020–22) and the post-period (2023–25) per standard deviation of exposure to natural language processing, with country fixed effects. Cells: country × three-digit occupation (job ads: at least ten job ads that state an experience requirement in every year; skill requirements and education requirements: at least ten entry-level ads and ten senior ads in every year), weighted by pre-period job ads. Columns (1) and (2): one exposure at a time; columns (3) and (4): both exposures in one regression; column (5): column (1) minus column (3), equal to column (4) times the slope of augmentation exposure on automation exposure (0.82). Augmentation exposure: Appendix B. Standard errors clustered by three-digit occupation in parentheses; *, ** and *** denote significance at 10, 5 and 1 percent.